

YAMAHA

MUSIC SYNTHESIZER

SY99

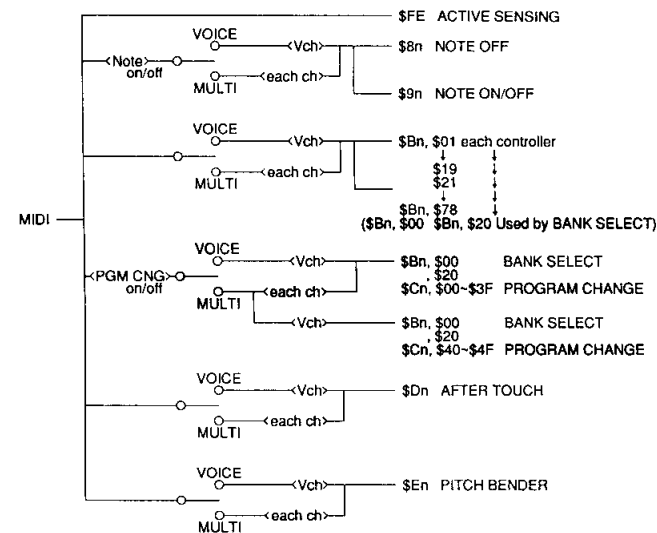
MIDI DATA FORMAT

1. Synthesizer mode

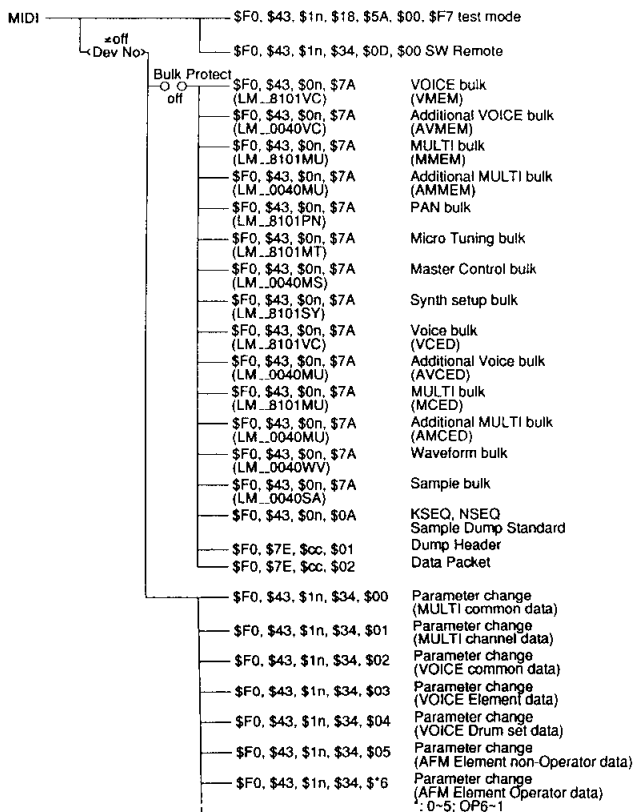
1.1 MIDI reception/transmission block diagram

< MIDI reception conditions > 1/2

Vch ... Voice Receive ch.

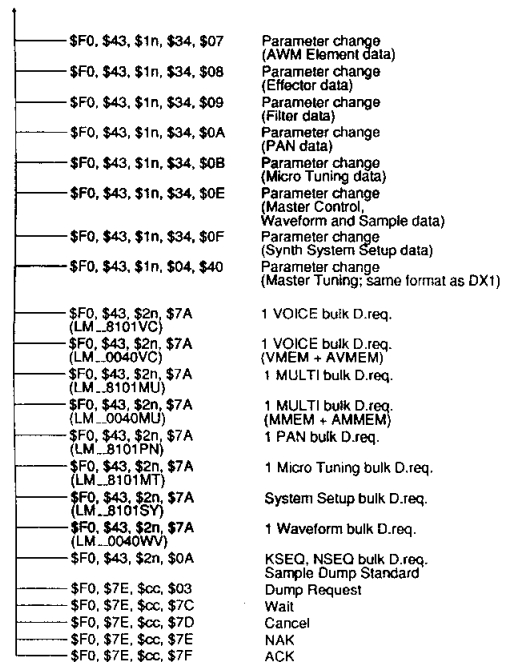


< MIDI reception conditions > 2/2



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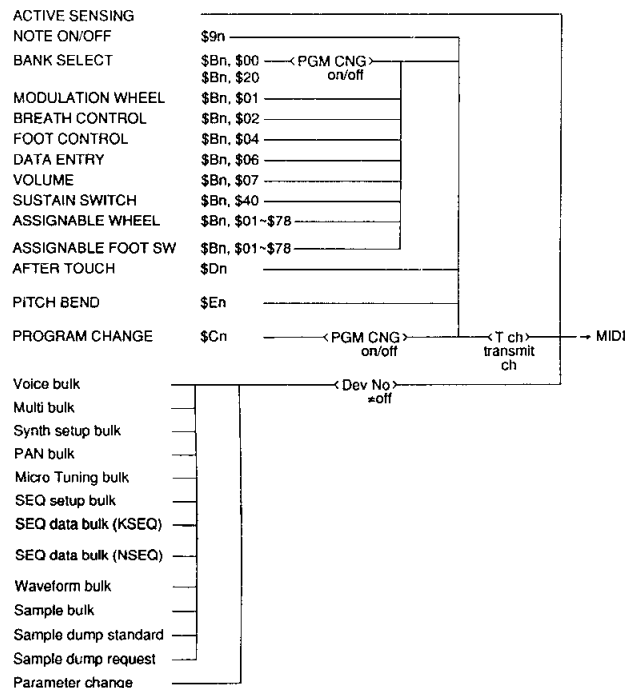
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Dev No = Device Number

VCED = Voice edit buffer
 AVCED = Additional Voice edit buffer
 MCED = Multi edit buffer
 AMCED = Additional Multi edit buffer
 VMEM = Voice memory
 AVMEM = Additional Voice memory
 MMEM = Multi memory
 AMMEM = Additional Multi memory

< MIDI transmission conditions >



1.2 Channel messages

1.2.1 Transmission

1.2.1.1 Note on/off

Transmission note range = E0 (28) ... G6 (103)
Velocity range = 0 ... 127 (0: note off)

1.2.1.2 Control change

When the following controllers are moved MIDI data will be transmitted.

ctrl#	parameter	data rng
1	Modulation wheel	0~127
2	Breath control	0~127
4	Foot control	0~127
6	Data entry slider	0~127
7	Volume pedal	0~127
64	Sustain switch	0, 127
96	data increment Sw.	127
97	data decrement Sw.	127
0~120	Assignable wheel	0~127
0~120	Assignable foot sw	0, 127

← Transmitted in all modes
other than Voice
Play mode or Multi Play
mode.

1.2.1.3 Program change

When a voice is selected in voice mode, or when a multi is selected in multi mode, a program change is transmitted. The program change number is assigned according to the mode, as follows.

		Bank No. (MSB), (LSB)	PGM CNG No.
Voice	Internal	1 (\$00), (\$00)	1~64 (\$00~\$3F)
	Card	2 (\$00), (\$01)	↓
	Preset 1	3 (\$00), (\$02)	↓
	Preset 2	6 (\$00), (\$05)	↓
Multi	Internal	17 (\$00), (\$10)	65~80 (\$40~\$4F)
	Card	18 (\$00), (\$11)	↓
	Preset	19 (\$00), (\$12)	↓
Voice In Multi	Internal	33 (\$00), (\$20)	1~64 (\$00~\$3F)
	Card	34 (\$00), (\$21)	↓
	Preset 1	35 (\$00), (\$22)	↓
	Preset 2	38 (\$00), (\$25)	↓

BANK SELECT \$Bn, \$00, \$00~\$7F (MSB)
 \$Bn, \$20, \$00~\$7F (LSB)
PROGRAM CHANGE \$Cn, \$00~\$7F

No bank select or program change messages are transmitted if Program Change mode is set to OFF.

1.2.1.4 Pitch bend

Pitch bend is transmitted with 7 bit precision.

1.2.1.5 Aftertouch

Transmitted from MIDI when aftertouch is operated.

1.2.1.6 Channel mode messages

Channel mode messages are not transmitted.

1.2.2 Reception

1.2.2.1 Note on/off

Received note range = C-2 (0) ... G8 (127)
Velocity range = 1 ... 127 (note on only)

* A system setup parameter allows selection of
normal = receive all note numbers
odd = receive odd note numbers
even = receive even note numbers

1.2.2.2 Control change

The following parameters can be controlled via MIDI.

cntrl#	parameter	data rng
1~120	Pitch Modulation	0...127
1~120	Amplitude Modulation	0...127
1~120	Filter Modulation	0...127
1~120	Pan LFO Modulation	0...127
1~120	Cutoff Bias	0...127
1~120	Pan Bias	0...127
1~120	EG Bias	0...127
1~120	Volume (assignable)	0...127
5	Portamento Time	0...127
7	Volume	0...127
64	Sustain Switch	0...127
65	Portamento Switch	0...127

Control #s 0(\$00) and 32(\$20) are used for bank select.

1.2.2.3 Program change

The SY99 responds as follows when a program change message is received:

- 1) If Program Change mode is OFF:
: Bank select and program change messages are not recognized.
- 2) If Program Change mode is ON:
: Response as described by the chart in 2.2.1.3. Voice selection is carried out upon reception of the program change message.

1.2.2.4 Pitch bend

Pitch bend is recognized by the MSB only.

1.2.2.5 Aftertouch

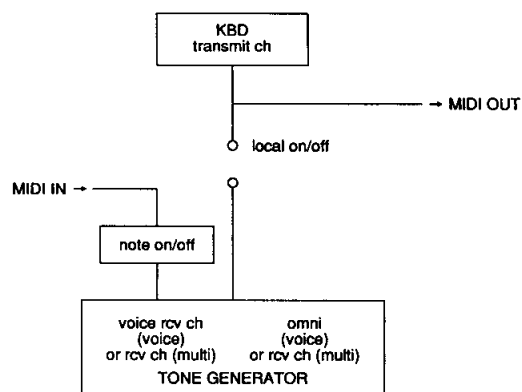
Channel aftertouch is recognized. Polyphonic aftertouch is not recognized.

1.2.2.6 Channel mode messages

Not received

1.2.3 Diagram of the keyboard section and tone generator section

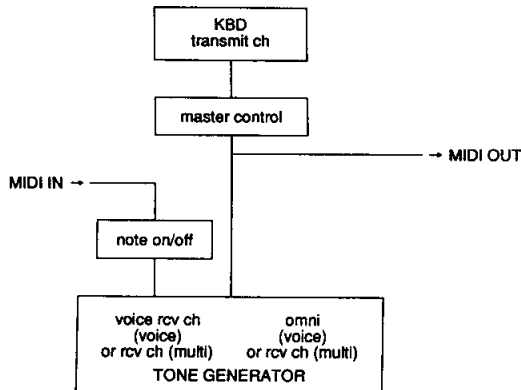
* Normal mode



Note 1) In voice mode, sound will be produced even if the Voice Receive Channel does not match the Keyboard Trans. Channel.

Note 2) Notes from the keyboard and notes from MIDI will be distinguished in polyphonic mode, and not distinguished in monophonic mode. Controller data from the keyboard and controller data from the MIDI (control change, aftertouch, pitch bend) will not be distinguished except for the sustain switch.

* Master control mode



2. System exclusive messages

2.1 Parameter changes

The SY99 is capable of transmitting and recognizing the following 17 types of parameter change message (with the exception of 14. Switch Remote, which is recognized but cannot be transmitted). Recognition of 14. Switch Remote has the same effect on the display as pressing the switch.

- 1). Multi Common Data parameter change
- 2). Multi Channel Data parameter change
- 3). Voice Common Data parameter change
- 4). Voice Element Data parameter change
- 5). Voice Drum Set Data parameter change
- 6). AFM Element Common Data parameter change
- 7). AFM Element Operator Data parameter change
- 8). AWM Element Data parameter change
- 9). Waveform Data parameter change
- 10). Effect Data parameter change
- 11). Filter Data parameter change
- 12). PAN Data parameter change
- 13). Micro Tuning Data parameter change
- 14). Switch Remote parameter change
- 15). Master Control Data parameter change
- 16). System Setup Data parameter change
- 17). Sample Data parameter change

Parameter message recognition can be turned off by turning the Device Number switch off. It cannot be turned off using any other MIDI switch.

2.1.1 Multi Common Data parameter change

```

11110000 F0
01000011 43
0001nnnn nnnn - Device Number
00110100 34
00000000 00
00000000 00
00000000 00
00000000 00
000ppppp ppppp - Table 1-1, N2
00000000 00
0vvvvvvv vvvvvvv - ascii
11110111 F7
  
```

This message modifies Multi Common data (multi name) one parameter at a time. When this message is received, the following will occur.

Voice Play mode : ignore.
 Voice Edit mode : ignore.
 Multi Play mode : move to Multi Edit mode and receive. (Display changes.)
 Multi Edit mode : receive. (Display changes.)
 Voice Edit mode in Multi : receive.
 Utility mode (Voice mode) : ignore.
 Utility mode (Multi mode) : move to Voice Edit mode and receive. (Display changes.)

2.1.2 Multi Channel Data parameter change

```

11110000 F0
01000011 43
0001nnnn nnnn - Device Number
00110100 34
00000001 01
0000cccc cccc - Voice Channel Number
00000000 00
000ppppp ppppp - Table 1-2, N2
00000000 00
0vvvvvvv vvvvvvv - Data Value
11110111 F7
  
```

This message modifies the channel data of a multi one parameter at a time. When this message is received, the following will occur.

Voice Play mode : ignore.
 Voice Edit mode : ignore.
 Multi Play mode : move to Multi Edit mode and receive. (Display changes.)
 Multi Edit mode : receive. (Display changes.)
 Voice Edit mode in Multi : receive.
 Utility mode (Voice mode) : ignore.
 Utility mode (Multi mode) : move to Voice Edit mode and receive. (Display changes.)

2.1.3 Voice Common Data parameter change

```

11110000 F0
01000011 43
0001nnnn nnnn - Device Number
00110100 34
00000010 02
00000000 00
00000000 00
00000000 00
0ppppppp pppppp - Table 1-3, N2
00000000 00
0vvvvvvv vvvvvvv - Data Value
11110111 F7
  
```

This message modifies voice common data one parameter at a time. When this message is received, the following will occur.

Voice Play mode : move to Voice Edit mode and receive. (Display changes.)
 Voice Edit mode : receive. (Display changes.)
 Multi Play mode : ignore.
 Multi Edit mode : ignore.
 Voice Edit mode in Multi : receive. (Display changes.)
 Utility mode (Voice mode) : move to Voice Edit mode and receive. (Display changes.)
 Utility mode (Multi mode) : ignore.

2.1.4 Voice Element Data parameter change

```

11110000 F0
01000011 43
0001nnnn nnnn - Device Number
00110100 34
00000011 03
0ee00000 ee - Element Number
00000000 00
0000pppp pppp - Table 1-4, N2
00000000 00
0vvvvvvv vvvvvvv - Data Value
11110111 F7
  
```

This message modifies voice element data one parameter at a time. When this message is received, the following will occur.

Voice Play mode : if the specified element exists, move to Voice Edit mode and receive. If not, ignore.
 Voice Edit mode : if the specified element exists, receive. if not, ignore. (The screen goes into Edit Mode.)
 Multi Play mode : ignore.
 Multi Edit mode : ignore.
 Voice Edit mode in Multi : same as for Voice Edit mode.
 Utility mode (Voice mode) : same as for Voice Play mode.
 Utility mode (Multi mode) : ignore.

2.1.5 Voice Drum Set Data parameter change

```

11110000 F0
01000011 43
0001nnnn nnnn - Device Number
00110100 34
00000100 04
0mmmmmm mmmmmm - MIDI Note Number
00000000 00
0000pppp pppp - Table 1-5, N2
0000000v v - Data Value (MSB)
0vvvvvvv vvvvvvv - Data Value (LS7bits)
11110111 F7

```

This message modifies voice drum set data one parameter at a time. When this message is received, the following will occur.

Voice Play mode : if originally a drum set, move to Voice Edit mode and receive. (Display changes.) If not, ignore. (The screen goes into Edit Mode.)

Voice Edit mode : if originally a drum set, receive. (Display changes.) If not, ignore.

Multi Play mode : ignore.

Multi Edit mode : ignore.

Voice Edit mode in Multi : same as for Voice Edit mode.

Utility mode (Voice mode) : same as for Voice Play mode.

Utility mode (Multi mode) : ignore.

2.1.6 AFM Element Common Data parameter change

```

11110000 F0
01000011 43
0001nnnn nnnn - Device Number
00110100 34
00000101 05
0ee00000 ee - Element Number
00000000 00
000ppppp ppppp - Table 1-6, N2
00000000 00
0vvvvvvv vvvvvvv - Data Value
11110111 F7

```

This message modifies AFM element common data one parameter at a time. When this message is received, the following will occur.

Voice Play mode : if the specified element exists and is AFM, move to Voice Edit mode and receive. (Display changes.) If not, ignore. (The screen goes into Edit Mode.)

Voice Edit mode : if the specified element exists and is AFM, receive. If not, ignore. (Display changes.)

Multi Play mode : ignore.

Multi Edit mode : ignore.

Voice Edit mode in Multi : same as Voice Edit mode.

Utility mode (Voice mode) : same as Voice Play mode.

Utility mode (Multi mode) : ignore.

2.1.7 AFM Element Operator Enable Parameter change

```

11110000 F0
01000011 43
0001nnnn nnnn - Device Number
00110100 34
00000101 05
0ee00000 ee - Element Number
01111111 7F
01111111 7F
00000000 00
0vvvvvvv vvvvvvv - Data Value
11110111 F7

```

This message modifies AFM operator enable data one parameter at a time. When this message is received, the following will occur.

Voice Play mode : if the specified element exists and is AFM, move to Voice Edit mode and receive. (Display changes.) If not, ignore. (The screen goes into Edit Mode.)

Voice Edit mode : if the specified element exists and is AFM, receive. If not, ignore. (Display changes.)

Multi Play mode : ignore.

Multi Edit mode : ignore.

Voice Edit mode in Multi : same as Voice Edit mode.

Utility mode (Voice mode) : same as Voice Play mode.

Utility mode (Multi mode) : ignore.

2.1.8 AFM Element Operator Data Parameter change

```

11110000 F0
01000011 43
0001nnnn nnnn - Device Number
00110100 34
0ooo0110 ooo - 6-(Operator Number)
0ee00000 ee - Element Number
00000000 00
00pppppp pppppp - Table 1-7, N2
0000000v v - Data Value (MSB)
0vvvvvvv vvvvvvv - Data Value (LS7bits)
11110111 F7

```

This message modifies AFM operator parameter data one parameter at a time. When this message is received, the following will occur.

Voice Play mode : if the specified element exists and is AFM, move to Voice Edit mode and receive. (Display changes.) If not, ignore. (The screen goes into Edit Mode.)

Voice Edit mode : if the specified element exists and is AFM, receive. (Display changes.) If not, ignore.

Multi Play mode : ignore.

Multi Edit mode : ignore.

Voice Edit mode in Multi : same as in Voice Edit mode.

Utility mode (Voice mode) : same as in Voice Play mode.

Utility mode (Multi mode) : ignore.

2.1.9 AWM Element Data parameter change

```

11110000 F0
01000011 43
0001nnnn nnnn - Device Number
00110100 34
00000111 07
0ee00000 ee - Element Number
00000000 00
0ppppppp ppppppp - Table 1-8, N2
0000000v v - Data Value (MSB)
0vvvvvvv vvvvvvv - Data Value (LS7bits)
11110111 F7

```

This message modifies AWM element data one parameter at a time. When this message is received, the following will occur.

Voice Play mode : if the specified element exists and is AWM, move to Voice Edit mode and receive. (Display changes.) If not, ignore. (The screen goes into Edit Mode.)

Voice Edit mode : if the specified element exists and is AWM, receive. If not, ignore. (Display changes.)

Multi Play mode : ignore.

Multi Edit mode : ignore.

Voice Edit mode in Multi : same as in Voice Edit mode.

Utility mode (Voice mode) : same as in Voice Play mode.

Utility mode (Multi mode) : ignore.

2.1.10 Waveform and Sample parameter change

Waveform parameter change

```

11110000 F0
01000011 43
0001nnnn nnnn - Device Number
00110100 34
00001110 0E
00wwwwww wwwwww - Waveform Number
00000101 05
0ppppppp ppppppp - Table 1-9, N2
00000000 00
0vvvvvvv vvvvvvv - Data Value
11110111 F7

```

Sample parameter change

```

11110000 F0
01000011 43
0001nnnn nnnn - Device Number
00110100 34
00001110 0E
0sssssss ssssss - Sample Number
00000101 06
0ppppppp ppppppp - Table 1-17, N2
00000000 00
0vvvvvvv vvvvvvv - Data Value
11110111 F7

```

These messages change a single parameter of waveform or sample data. These messages are recognized, and the data for the specified waveform or sample are changed, regardless of the SY99's current mode setting.

2.1.11 Effect Data parameter change

```

11110000 F0
01000011 43
0001nnnn nnnn - Device Number
00110100 34
00001000 08
00000000 00
00000000 00
000ppppp ppppp - Table 1-10, N2
00000000 00
0vvvvvvv vvvvvv - Data Value
11110111 F7

```

This message modifies effect data one parameter at a time. When this message is received, the following will occur.

Voice Play mode : move to Voice Edit mode and receive.
Voice Edit mode : receive. (Display changes.)
Multi Play mode : move to Multi Edit mode and receive.
Multi Edit mode : receive. (Display changes.)
Voice Edit mode in Multi : receive. Modify the Multi effect. (Display changes.)
Utility mode (Voice mode) : same as in Voice Play mode.
Utility mode (Multi mode) : same as in Multi Play mode.

2.1.12 Filter Data parameter change

```

11110000 F0
01000011 43
0001nnnn nnnn - Device Number
00110100 34
00001001 09
0ee00fff ee - Element Number, fff = filter Number
00000000 00
00pppppp ppppp - Table 1-11, N2
0000000v v - Data Value (MSB)
0vvvvvvv vvvvvv - Data Value (LS7bits)
11110111 F7

```

This message modifies filter data one parameter at a time. When this message is received, the following will occur.

Voice Play mode : if the specified element exists, move to Voice Edit mode and receive. If not, ignore.
Voice Edit mode : if the specified element exist, receive. If not, ignore. (Display changes.)
Multi Play mode : ignore.
Multi Edit mode : ignore.
Voice Edit mode in Multi : same as in Voice Edit mode.
Utility mode (Voice mode) : same as in Voice Play mode.
Utility mode (Multi mode) : ignore.

2.1.13 PAN Data parameter change

```

11110000 F0
01000011 43
0001nnnn nnnn - Device Number
00110100 34
00001010 0A
000mmmmm mmmmm - Memory Number
00000000 00
000ppppp ppppp - Table 1-12, N2
00000000 00
0vvvvvvv vvvvvv - Data Value
11110111 F7

```

This message modifies dynamic pan data one parameter at a time. When this message is received, the mode will not change, and the pan data of the specified memory will be modified.

2.1.14 Micro Tuning Data parameter change

```

11110000 F0
01000011 43
0001nnnn nnnn - Device Number
00110100 34
00001010 0B
000mmmmm mmmmm - Memory Number
0000000p p - Table 1-13, N1
0ppppppp pppppp - Table 1-13, N2
0vvvvvvv vvvvvv - Data Value (MS7bits)
0vvvvvvv vvvvvv - Data Value (LS7bits)
11110111 F7

```

This message modifies micro tuning data one parameter at a time. When this message is received, the mode will not change, and the micro tuning data of the specified memory will be modified.

2.1.15 Switch Remote parameter change

```

11110000 F0
01000011 43
0001nnnn nnnn - Device Number
00110100 34
00001010 0D
00000000 00
00000000 00
0ppppppp pppppp - Table 1-14, N2
00000000 00
0vvvvvvv vvvvvv - Data Value
11110111 F7

```

This message is only received, and allows remote control of all panel switches. The message will have the same effect as if that switch had been pressed.

2.1.16 Master Control Data parameter change

```

11110000 F0
01000011 43
0001nnnn nnnn - Device Number
00110100 34
00001110 0E
00nn0ttt nnttt - Table 1-15, T2
0ppppppp pppppp - Table 1-15, N1
0ppppppp pppppp - Table 1-15, N2
0vvvvvvv vvvvvv - Data Value
0vvvvvvv vvvvvv - Data Value
11110111 F7

```

This message modifies master control data one parameter at a time. When this message is received, the mode will not change.

2.1.17 System Setup Data parameter change

```

11110000 F0
01000011 43
0001nnnn nnnn - Device Number
00110100 34
00001111 0F
00000000 00
00000000 00
0ppppppp pppppp - Table 1-16, N2
00000000 00
0vvvvvvv vvvvvv - Data Value
11110111 F7

```

This message modifies system setup data one parameter at a time. When this message is received, the mode will not change.

3. Bulk dump

Reception is possible any time when not playing back or recording. The data is transmitted when the MIDI UTILITY "bulk dump" or SAMPLE UTILITY "sample dump" is executed or when a dump request is received.

3.1 Voice data bulk dump

There are nine types of voice data bulk dump as follows.

- 1AFM Voice bulk dump
- 2AFM Voice bulk dump
- 4AFM Voice bulk dump
- 1AWM Voice bulk dump
- 2AWM Voice bulk dump
- 4AWM Voice bulk dump
- 1AFM_1AWM Voice bulk dump
- 2AFM_2AWM Voice bulk dump
- Drum_set Voice bulk dump

```

0 11110000 F0
1 01000011 43
2 0000nnnn nnnn - Device Number
3 01110101 7A
4 0bbbbbbb byte count
5 0bbbbbbb
6 01001100 4C(ascii"L")
7 01001101 4D(ascii"M")
8 00100000 20(ascii" ")
9 00100000 20(ascii" ")
10 00110000 38(ascii"8")
11 00110001 31(ascii"1")
12 00110000 30(ascii"0")
13 00110001 31(ascii"1")
14 01010110 56(ascii"V")
15 01000011 43(ascii"C")
16 00000000 00
↓
29 00000000 00
30 0ttttttt tttttt - Memory_type
31 00mmmmmm mmmmmm - Memory Number
32 0ddddd dddddd - data
↓
0sssssss ssssss - check_sum
11110111 F7

```

MIDI Utility 1 Voice Bulk transmission

Memory_type = 7F
Memory Number = 00

MIDI Utility 64 Voices Bulk or Vc & Mlt Bulk transmission

Memory_type = 00 (INT)
Memory Number = Voices 0~63 will be transmitted sequentially. (For Vc & Mlt Bulk transmission, Voices 0~63 will be transmitted, followed by Multi Bulk Data.)

1 Voice Bulk transmission by Dump Request

Memory_type = 00 (INT)
02 (PRESET1)
03 (PRESET2)
Memory Number = 0~63 (the number specified by the Dump Request)

For reception, data will be processed as Edit_buffer when the Memory_type is 7F, and as INTERNAL for other cases.

* For details of the bulk dump data and dump request format, refer to table 2-1.

For transmission at the main unit panel operation, additional voice data is also transmitted. When a dump request is received, only voice data is transmitted.

3.2 Additional voice data bulk dump

- 1AFM & 1AWM Voice bulk dump
- 2AFM & 2AWM & 1AFM_1AWM Voice bulk dump
- 4AFM & 4AWM & 2AFM_2AWM Voice bulk dump
- Drum_set Voice bulk dump

```
0 11110000 F0
1 01000011 43
2 0000nnnn nnnn - Device Number
3 01111010 7A
4 0bbbbbbb 7A byte count
5 0bbbbbbb
6 01001100 4C(ascii"L")
7 01001101 4D(ascii"M")
8 00100000 20(ascii" ")
9 00100000 20(ascii" ")
10 00110000 30(ascii"0")
11 00110000 30(ascii"0")
12 00110100 34(ascii"4")
13 00110000 30(ascii"0")
14 01010110 56(ascii"V")
15 01000011 43(ascii"C")
16 00000000 00
↓
29 00000000 00
30 0ttttttt tttttt - Memory_type
31 00mmmmmm mmmmmm - Memory Number
32 0ddddd dddddd - data
↓
0sssssss ssssss - check_sum
11110111 F7
```

MIDI Utility 1 Voice Bulk transmission

Memory_Type = 7F
Memory Number = 00

MIDI Utility 64 Voices Bulk or Vc & Mlt Bulk transmission

Memory_type = 00 (INT)
Memory Number = Voices 0~63 will be transmitted sequentially. (For Vc & Mlt Bulk transmission, Voices 0~63 will be transmitted, followed by Multi Bulk Data.)

1 Voice Bulk transmission by Dump Request

Memory_type = 00 (INT)
02 (PRESET1)
03 (PRESET2)
Memory Number = 0~63 (the number specified by the Dump Request)

For reception, data will be processed as Edit_buffer when the Memory_type is 7F, and as INTERNAL for other cases.

* For details of the bulk dump data and dump request format, refer to table 2-2.

For transmission at the main unit panel operation or at the reception of a dump request, voice data is also transmitted.

3.3 Multi data bulk dump

```
0 11110000 F0
1 01000011 43
2 0000nnnn nnnn - Device Number
3 01111010 7A
4 0bbbbbbb 7A byte count
5 0bbbbbbb
6 01001100 4C(ascii"L")
7 01001101 4D(ascii"M")
8 00100000 20(ascii" ")
9 00100000 20(ascii" ")
10 00110000 30(ascii"0")
11 00110001 31(ascii"1")
12 00110000 30(ascii"0")
13 00110001 31(ascii"1")
14 01001101 4D(ascii"M")
15 01010101 55(ascii"U")
16 00000000 00
↓
29 00000000 00
30 0ttttttt tttttt - Memory_type
31 00mmmmmm mmmmmm - Memory Number
32 0ddddd dddddd - data
↓
0sssssss ssssss - check_sum
11110111 F7
```

MIDI Utility 1 Multi Bulk transmission

Memory_type = 7F
Memory Number = 00

MIDI Utility 16 Multi Bulk or Vc & Mlt Bulk transmission

Memory_type = 00 (INT)
Memory Number = Multi 0~15 will be transmitted sequentially. (For Vc & Mlt Bulk transmission, Voices 0~63 will be transmitted, followed by Multi Bulk Data.)

1 Multi Bulk transmission by Dump Request

Memory_type = 00 (INT)
02 (PRESET)
Memory Number = 0~15 (the number specified by the Dump Request)

For reception, data will be processed as Edit_buffer when the Memory_type is 7F, and as INTERNAL for other cases.

* For details of the bulk dump data and dump request format, refer to table 3-1.

For transmission at the main unit panel operation, additional voice data is also transmitted. When a dump request is received, only multi data is transmitted.

3.4 Additional multi data bulk dump

```
0 11110000 F0
1 01000011 43
2 0000nnnn nnnn - Device Number
3 01111010 7A
4 0bbbbbbb 7A byte count
5 0bbbbbbb
6 01001100 4C(ascii"L")
7 01001101 4D(ascii"M")
8 00100000 20(ascii" ")
9 00100000 20(ascii" ")
10 00110000 30(ascii"0")
11 00110000 30(ascii"0")
12 00110100 34(ascii"4")
13 00110000 30(ascii"0")
14 01001101 4D(ascii"M")
15 01010101 55(ascii"U")
16 00000000 00
↓
29 00000000 00
30 0ttttttt tttttt - Memory_type
31 00mmmmmm mmmmmm - Memory Number
32 0ddddd dddddd - data
↓
0sssssss ssssss - check_sum
11110111 F7
```

MIDI Utility 1 Multi Bulk transmission

Memory_type = 7F
Memory Number = 00

MIDI Utility 16 Multi Bulk or Vc & Mlt Bulk transmission

Memory_type = 00 (INT)
Memory Number = Multi 0~15 will be transmitted sequentially. (For Vc & Mlt Bulk transmission, Voices 0~63 will be transmitted, followed by Multi Bulk Data.)

1 Multi Bulk transmission by Dump Request

Memory_type = 00 (INT) 02 (PRESET)
Memory Number = 0~15 (the number specified by the Dump Request)

For reception, data will be processed as Edit_buffer when the Memory_type is 7F, and as INTERNAL for other cases.

* For details of the bulk dump data and dump request format, refer to table 3-2.

For transmission at the main unit panel operation or at the reception of a dump request, voice data is also transmitted.

3.5 Pan data bulk dump

```

0 11110000 F0
1 01000011 43
2 0000nnnn nnnn - Device Number
3 01111010 7A
4 0bbbbbbb byte count
5 0bbbbbbb
6 01001100 4C(ascii"L")
7 01001101 4D(ascii"M")
8 00100000 20(ascii" ")
9 00100000 20(ascii" ")
10 00111000 38(ascii"8")
11 00110001 31(ascii"1")
12 00110000 30(ascii"0")
13 00110001 31(ascii"1")
14 01010000 50(ascii"P")
15 01001110 4E(ascii"N")
16 00000000 00
↓
29 00000000 00
30 0ttttttt tttttt - Memory_type
31 00mmmmmm mmmmmm - Memory Number
32 0ddddd dddddd - data
↓
0sssssss ssssss - check_sum
11110111 F7
  
```

MIDI Utility PAN Bulk transmission

Memory_type = 00 (INT)
Memory Number = start from 0 and transmit consecutively to 31

PAN Bulk transmission by Dump Request

Memory_type = 00 (INT) 02 (PRESET)
Memory Number = 0~63 (the number specified by the Dump Request)

For reception, bits 5 and 6 of Memory_type and memory Number will be ignored, and the data will be processed as INTERNAL.

* For details of the bulk dump data and dump request format, refer to table 4.

3.6 Micro tuning data bulk dump

```

0 11110000 F0
1 01000011 43
2 0000nnnn nnnn - Device Number
3 01111010 7A
4 0bbbbbbb byte count
5 0bbbbbbb
6 01001100 4C(ascii"L")
7 01001101 4D(ascii"M")
8 00100000 20(ascii" ")
9 00100000 20(ascii" ")
10 00111000 38(ascii"8")
11 00110001 31(ascii"1")
12 00110000 30(ascii"0")
13 00110001 31(ascii"1")
14 01001101 4D(ascii"M")
15 01010100 54(ascii"T")
16 00000000 00
↓
29 00000000 00
30 00000000 00 - Memory_type
31 00mmmmmm mmmmmm - Memory Number
32 0ddddd dddddd - data
↓
0sssssss ssssss - check_sum
11110111 F7
  
```

MIDI Utility Micro Tuning Bulk transmission

Memory_type = 00 (INT)
Memory Number = start from 0 and transmit consecutively to 1

MCT Bulk transmission by Dump Request

Memory_type = 00 (INT)
Memory Number = 0~63 (the number specified by the Dump Request)

For reception, bits 1~6 of Memory_type and Memory Number will be ignored, and the data will be processed as INTERNAL.

* For details of the bulk dump data and dump request format, refer to table 5.

3.7 Master control data bulk dump

```

0 11110000 F0
1 01000011 43
2 0000nnnn nnnn - Device Number
3 01111010 7A
4 0bbbbbbb byte count
5 0bbbbbbb
6 01001100 4C(ascii"L")
7 01001101 4D(ascii"M")
8 00100000 20(ascii" ")
9 00100000 20(ascii" ")
10 00111000 30(ascii"0")
11 00110000 30(ascii"0")
12 00110100 34(ascii"4")
13 00110000 30(ascii"0")
14 01001101 4D(ascii"M")
15 01010011 53(ascii"S")
16 00000000 00
↓
31 00000000 00
32 0ddddd dddddd - data
↓
0sssssss ssssss - check_sum
11110111 F7
  
```

* For details of the bulk dump data and dump request format, refer to table 6.

Data transmitted at the main unit panel operation is followed by SYNTH system setup data when it is transmitted.

For transmission at the reception of a dump request, only master control data is transmitted.

3.8 SYNTH system setup data bulk dump

```

0 11110000 F0
1 01000011 43
2 0000nnnn nnnn - Device Number
3 01111010 7A
4 0bbbbbbb byte count
5 0bbbbbbb
6 01001100 4C(ascii"L")
7 01001101 4D(ascii"M")
8 00100000 20(ascii" ")
9 00100000 20(ascii" ")
10 00111000 38(ascii"8")
11 00110001 31(ascii"1")
12 00110000 30(ascii"0")
13 00110001 31(ascii"1")
14 01010011 53(ascii"S")
15 01011001 59(ascii"Y")
16 00000000 00
↓
31 00000000 00
32 0ddddd dddddd - data
↓
0sssssss ssssss - check_sum
11110111 F7
  
```

* For details of the bulk dump data and dump request format, refer to table 7.

For transmission at the main unit panel operation, master control data is transmitted before this data.

For transmission at the reception of a dump request, only this data is transmitted.

3.9 SEQ setup data bulk dump

```

0 11110000 F0
1 01000011 43
2 0000nnnn nnnn - Device Number
3 01111110 7E
4 0bbbbbbb byte count
5 0bbbbbbb
6 01001100 4C(ascii"L")
7 01001101 4D(ascii"M")
8 00100000 20(ascii" ")
9 00100000 20(ascii" ")
10 00110000 38(ascii"8")
11 00110001 31(ascii"1")
12 00110000 30(ascii"0")
13 00110001 31(ascii"1")
14 01010011 53(ascii"S")
15 01010011 53(ascii"S")
16 0ddddd - data
    ↓
    0sssssss ssssss - check_sum
    11110111 F7
  
```

* For details of the bulk dump data, refer to table 8-1.

Dump request

```

0 11110000 F0
1 01000011 43
2 0000nnnn nnnn - Device Number
3 01111110 7E
4 01001100 4C(ascii"L")
5 01001101 4D(ascii"M")
6 00100000 20(ascii" ")
7 00100000 20(ascii" ")
8 00110000 38(ascii"8")
9 00110001 31(ascii"1")
10 00110000 30(ascii"0")
11 00110001 31(ascii"1")
12 01010011 53(ascii"S")
13 01010011 53(ascii"S")
14 11110111 F7
  
```

3.10 SEQ song and pattern data bulk dump (KSEQ)

Reception is possible at any time when not playing or recording. This data is transmitted when the MIDI UTILITY "bulk dump" is executed, or when a dump request is received.

```

0 11110000 F0
1 01000011 43
2 0000nnnn nnnn - Device Number
3 00001010 0A
4 0bbbbbbb byte count
5 0bbbbbbb max. 4096
6 01001100 4C(ascii"L")
7 01001101 4D(ascii"M")
8 00100000 20(ascii" ")
9 00100000 20(ascii" ")
10 01001011 4B(ascii"K")
11 01010011 53(ascii"S")
12 01000101 45(ascii"E")
13 01010010 51(ascii"Q")
14 00100000 20(ascii" ")
15 00100000 20(ascii" ")
16 0ddddd - data
    ↓
    0sssssss ssssss - check_sum
    11110111 F7
  
```

When the byte count is below 4096, the byte count is that count. When the byte count exceeds 4096, a byte count and check_sum operation is repeated for every 4096 bytes marked off from the top.

Dump request

```

0 11110000 F0
1 01000011 43
2 0000nnnn nnnn - Device Number
3 00001010 0A
4 01001100 4C(ascii"L")
5 01001101 4D(ascii"M")
6 00100000 20(ascii" ")
7 00100000 20(ascii" ")
8 01001011 4B(ascii"K")
9 01010011 53(ascii"S")
10 01000101 45(ascii"E")
11 01010010 51(ascii"Q")
12 00100000 20(ascii" ")
13 00100000 20(ascii" ")
14 11110111 F7
  
```

3.11 SEQ song and pattern data bulk dump (NSEQ)

Reception is possible at any time when not playing or recording. This data is transmitted when the MIDI UTILITY "bulk dump" is executed, or when a dump request is received.

```

0 11110000 F0
1 01000011 43
2 0000nnnn nnnn - Device Number
3 00001010 0A
4 0bbbbbbb byte count
5 0bbbbbbb max. 4096
6 01001100 4C(ascii"L")
7 01001101 4D(ascii"M")
8 00100000 20(ascii" ")
9 00100000 20(ascii" ")
10 01001110 4E(ascii"N")
11 01010011 53(ascii"S")
12 01000101 45(ascii"E")
13 01010010 51(ascii"Q")
14 00100000 20(ascii" ")
15 00100000 20(ascii" ")
16 0ddddd - data
    ↓
    0sssssss ssssss - check_sum
    11110111 F7
  
```

When the byte count is below 4096, the byte count is that count. When the byte count exceeds 4096, a byte count and check_sum operation is repeated for every 4096 bytes marked off from the top.

Dump request

```

0 11110000 F0
1 01000011 43
2 0000nnnn nnnn - Device Number
3 00001010 0A
4 01001100 4C(ascii"L")
5 01001101 4D(ascii"M")
6 00100000 20(ascii" ")
7 00100000 20(ascii" ")
8 01001110 4E(ascii"N")
9 01010011 53(ascii"S")
10 01000101 45(ascii"E")
11 01010010 51(ascii"Q")
12 00100000 20(ascii" ")
13 00100000 20(ascii" ")
14 11110111 F7
  
```

3.12 Waveform data bulk dump

```

0 11110000 F0
1 01000011 43
2 0000nnnn nnnn - Device Number
3 01111010 7A
4 0bbbbbbb byte count
5 0bbbbbbb
6 01001100 4C(ascii"L")
7 01001101 4D(ascii"M")
8 00100000 20(ascii" ")
9 00100000 20(ascii" ")
10 00110000 30(ascii"0")
11 00110000 30(ascii"0")
12 00110100 34(ascii"4")
13 00110000 30(ascii"0")
14 01010111 57(ascii"W")
15 01010110 56(ascii"V")
16 00000000 00
    ↓
29 00000000 00
30 00000ttt ttt - Memory_type
31 0mmmmmmm mmmmmm - Memory Number
32 0ddddd - data
    ↓
    0sssssss ssssss - check_sum
    11110111 F7
  
```

Dump request

```

0 11110000 F0
1 01000011 43
2 0010nnnn nnnn - Device Number
3 01111010 7A
4 01001100 4C(ascii"L")
5 01001101 4D(ascii"M")
6 00100000 20(ascii" ")
7 00100000 20(ascii" ")
8 00110000 30(ascii"0")
9 00110000 30(ascii"0")
10 00110100 34(ascii"4")
11 00110000 30(ascii"0")
12 01010111 57(ascii"W")
13 01010110 56(ascii"V")
14 00000000 00
    ↓
27 00000000 00
28 00000ttt ttt - Memory_type
29 0mmmmmmm mmmmmm - Memory Number
30 11110111 F7
  
```

The Sample utility Waveform Bulk Dump operation transmits the following information:

Memory_type = \$00 (INT)
Memory number = \$00 to \$3F, in sequence

A bulk dump in response to a dump request message transmits the following:

Memory_type = \$00 (INT), \$01 (CARD), \$02, \$03 (PRE1), \$04 (PRE2)
Memory number = a number from \$00 to \$3F, as specified by the dump request

	Preset 1, numbers 1 to 128	Preset 1, numbers 129 to 155
Memory_type	\$02	\$03
Memory Number	\$00-\$7F	\$00-\$1A

Requests for nonexistent numbers are ignored.

Incoming bulk dumps are handled as INTERNAL memory; bit 6 of Memory_type and Memory Number are ignored.

* Refer to attached Table 9-1 for details regarding bulk dump data format.

4. Sample dumps

The SY99 is capable of handling two sample dumps in two formats: the Sample Dump Standard, and the SY99 Sample Bulk Dump. Sample dumps in both of these formats are recognized. Sample dumps in both are transmitted when the Sample utility Sample Dump operation is executed. When a Sample Dump Standard Request or an SY99 Sample Bulk Dump Request is received, each data is transmitted. The upper limit on sample numbers for dumps in either format is set at \$62; sample numbers higher than \$62 are treated as \$62.

Sample Dump Standard

```
DUMP REQ  F0, 7E, CC, 03, SS, SS, F7
ACK       F0, 7E, CC, 7F, PP, F7
NAK       F0, 7E, CC, 7E, PP, F7
CANCEL    F0, 7E, CC, 7D, PP, F7
WAIT      F0, 7E, CC, 7C, PP, F7
DATA PACKET F0, 7E, CC, 02, KK, <120 byte>, 11, F7
DUMP HEADER F0, 7E, CC, 01, SS, SS, EE, FF, FF, FF, GG, GG, GG,
             hh, hh, hh, ii, ii, ii, jj, F7
```

```
pp      : packet number
cc      : channel number
ss ss   : sample number (LSB first)
ee      : sample format (SY99 handles samples of
             8 to 16 bits)
ff ff ff : sample period (LSB first)
gg gg gg : sample length (LSB first)
hh hh hh : loop start (LSB first)
ii ii ii : loop end (LSB first)
jj      : loop type
kk      : running packet count (0-127)
             (number of current packet)
11      : checksum (XOR of 7E cc 02 kk <120 bytes>)
```

SY99 Sample Bulk Dump

```
0 11110000 F0
1 01000011 43
2 0000nnnn nnnn - Device Number
3 01111010 7A
4 0bbbbbbb byte count
5 0bbbbbbb
6 01001100 4C(ascii"L")
7 01001101 4D(ascii"M")
8 00100000 20(ascii" ")
9 00100000 20(ascii" ")
10 00110000 30(ascii"0")
11 00110000 30(ascii"0")
12 00110100 34(ascii"4")
13 00110000 30(ascii"0")
14 01010011 53(ascii"S")
15 01000001 41(ascii"A")
16 00000000 00
↓
30 00000000 00
31 0mmmmmmmm mmmmmmm - Memory number
32 0ddddd - data
↓
0sssssss
11110111 ssssss - check_sum
F7
```

Dump request

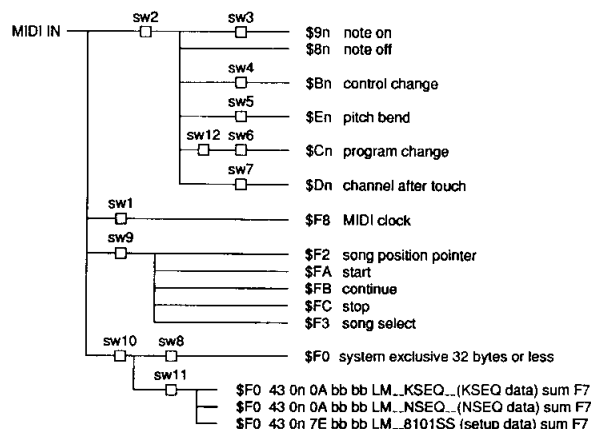
```
0 11110000 F0
1 01000011 43
2 0010nnnn nnnn - Device Number
3 01111010 7A
4 01001100 4C(ascii"L")
5 01001101 4D(ascii"M")
6 00100000 20(ascii" ")
7 00100000 20(ascii" ")
8 00110000 30(ascii"0")
9 00110000 30(ascii"0")
10 00110100 34(ascii"4")
11 00110000 30(ascii"0")
12 01010011 53(ascii"S")
13 01000001 41(ascii"A")
14 00000000 00
↓
28 00000000 00
29 0mmmmmmmm mmmmmmm - Memory number
30 11110111 F7
```

* Refer to attached Table 9-2 for details regarding bulk dump data format.

5. Sequencer mode

5.1 MIDI reception/transmission block diagram

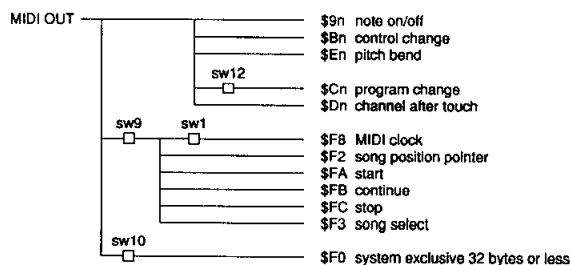
(Receive flow chart)



(Note)

- sw1 ☐ on when SEQ SETUP sync is set to MIDI
- sw2 ☐ during RECORD, the SEQ SETUP reception channel
- sw3 ☐ during RECORD, the SEQ SETUP velocity on/off
- sw4 ☐ during RECORD, the SEQ SETUP control change on/off
- sw5 ☐ during RECORD, the SEQ SETUP pitch bend on/off
- sw6 ☐ during RECORD, the SEQ SETUP program change on/off
- sw7 ☐ during RECORD, the SEQ SETUP channel pressure on/off
- sw8 ☐ during RECORD, the SEQ SETUP system exclusive (of 32 bytes or less) on/off
- sw9 ☐ on/off in SEQ SETUP for song position pointer, start, continue, stop
- sw10 ☐ set by the utility Device Number
- sw11 ☐ received only when utility Bulk Memory Protect = off
- sw12 ☐ received only when utility Program Change Mode is not off

(Transmit flow chart)



(Note)

- sw1 ☐ on when SEQ SETUP sync is set to INTERNAL
- sw9 ☐ on/off in SEQ SETUP for MIDI clock song position pointer, start, continue, stop
- sw10 ☐ set by the utility Device Number
- sw12 ☐ received only when utility Program Change Mode is not off

5.2 Channel messages

Received only during RECORD. Transmitted only during PLAY and overdubbing.

For transmission and reception conditions, refer to the Receive flow chart and Transmit flow chart.

5.3 Mode messages

Neither received nor transmitted.

5.4 System common messages

Receive only \$F2 and \$F7. All others are neither received nor transmitted.

5.5 System realtime messages

5.5.1 Status F8, FA, FB, FC

Receive.

5.5.2 Status F9, FD, FF

After decoding, do nothing.

6. Status FE (active sensing)

a) Transmission

Transmit FE at intervals of approximately 170 msec.

b) Reception

Once FE has been received, if no MIDI data arrives for an interval longer than approximately 300 msec, the MIDI receive buffer is cleared, and if any keys remain on, they will be turned off.

< Table 1-1 >

MIDI Parameter Change table (Multi mode Common Data)

\$F0, \$43, \$1n, \$34, \$00, \$00, \$00, N2, \$00, V2, \$F7

Note) n ; Device Number

V2 ; parameter value

[header section]

	N2	data name	data range	note
0	00	MNAM 0	ascii	Multi Voice Set Name
1	01	MNAM 1	ascii	"
2	02	MNAM 2	ascii	"
3	03	MNAM 3	ascii	"
4	04	MNAM 4	ascii	"
5	05	MNAM 5	ascii	"
6	06	MNAM 6	ascii	"
7	07	MNAM 7	ascii	"
8	08	MNAM 8	ascii	"
9	09	MNAM 9	ascii	"
10	0A	MNAM10	ascii	"
11	0B	MNAM11	ascii	"
12	0C	MNAM12	ascii	"
13	0D	MNAM13	ascii	"
14	0E	MNAM14	ascii	"
15	0F	MNAM15	ascii	"
16	10	MNAM16	ascii	"
17	11	MNAM17	ascii	"
18	12	MNAM18	ascii	"
19	13	MNAM19	ascii	"

< Table 1-2 >

MIDI Parameter Change table (Multi mode Channel Data)

\$F0, \$43, \$1n, \$34, \$01, T2, \$00, N2, \$00, V2, \$F7

Note) n ; Device Number

T2 ; voice channel number

V2 ; parameter value

[channel data]

	N2	data name	data range	note
0	00	OFVCSW INDV OUT0CH OUT1CH	b6 : off/on b5~2 : off, 1~8 b0 : off/on b1 : off/on	Channel 1 Off_voice_switch Individual Output select Output 1 select Output 2 select Note: INDV0 TG77 only valid.
1	01	VMEM	int, p1, p2, crd	Voice Memory Select
2	02	VNUM	0~63	Voice number
3	03	VOL	0~127	Volume
4	04	MTUN	-64~+63 (o/b)	Tuning
5	05	MNSF	-64~+63 (o/b)	Note shift
6	06	STPAN	voice, -31~+31 (o/b)	Performance static PAN Note: Voice PAN setting not valid if a setting other than "voice" is selected.
7	07	EFSDMD	voice/multi	effect send mode
8	08	EFLN1CH EFLN2CH EFLN3CH EFLN4CH EFSDLV	b0 : off/on b1 : off/on b2 : off/on b3 : off/on 0~127	effect send line 1 select effect send line 2 select effect send line 3 select effect send line 4 select effect send level
9	09			

Note) (o/b) ; offset binary

< Table 1-3 >

MIDI Parameter Change table (Voice data common data)

\$F0, \$43, \$1n, \$34, \$02, \$00, \$00, N2, \$00, V2, \$F7

Note) n ; channel number

V2 ; parameter value

[header data]

	N2	data name	data range	note
0	00	ELMODE	0: 1AFM_mono 1: 2AFM_mono 2: 4AFM_mono 3: 1AFM_poly 4: 2AFM_poly 5: 1AWM_poly 6: 2AWM_poly 7: 4AWM_poly 8: 1AFM_1AWM_poly 9: 2FM_2PCM_poly 10: DRUM_SET	—Element Select Mode—
1	01	VNAM0	ascii	—Voice Name— *
2	02	VNAM1	ascii	
3	03	VNAM2	ascii	
4	04	VNAM3	ascii	
5	05	VNAM4	ascii	
6	06	VNAM5	ascii	
7	07	VNAM6	ascii	
8	08	VNAM7	ascii	
9	09	VNAM8	ascii	
10	0A	VNAM9	ascii	

[Controllers]

	N2	data name	data range	note
11	28	WPBR	0~12	—Pitch Bend—
12	29	ATPBR	-12~+12 (s/m)	Wheel Pitch Bend Range
13	2A	PMASN	0~121	—Pitch Modulation—
14	2B	PMRNG	0~127	Device Assign (MIDI control #) Modulation range
15	2C	AMASN	0~121	—Amplitude Modulation—
16	2D	AMRNG	0~127	Device Assign (MIDI control #) Modulation range
17	2E	FMASN	0~121	—Filter Modulation—
18	2F	FMRNG	0~127	Device Assign (MIDI control #) Modulation range
19	30	PNLASN	0~121	—PAN Modulation— Note) Valid only when Multi is selected
20	31	PNLRNG	0~127	Device Assign (MIDI control #) Modulation range
21	32	COASN	0~121	—Filter Cut_off Bias—
22	33	CORNG	0~127	Device Assign (MIDI control #) Cut_off range
23	34	PNBASN	0~121	—PAN Bias— Note) Valid only when Multi is selected
24	35	PNBRNG	0~127	Device Assign (MIDI control #) Bias range
25	36	EGBASN	0~121	—EG Bias—
26	37	EGBRNG	0~127	Device Assign (MIDI control #) Bias range
27	38	VVLASN	0~121	—Voice Volume—
28	39	VVLLML	0~127	Device Assign (MIDI control #) Volume Limit Low

Note) For the above Device Assigns, 121 will select After Touch.

[Only for Normal]

	N2	data name	data range	note
29	3A	MCTUN	0~65	Micro Tuning table select
30	3B	RNDP	0~7	Random Pitch fluctuation
31	3C	PORM	0, 1	—Portamento— Note: Only valid in FM_Element only Voice modes.
32	3D	POS	0~127	Mode Time
33	3E	reserve		
34	3F	VVOL	0~127	Voice Volume
35	40	reserve		
36	41	reserve		
37	42	AFTMD	all, top, btm, hi, low	zoned after touch mode
38	43	SPTPNT	0~127	zoned after touch split point

Note) (s/m) ; sign magnitude

< Table 1-4 >

MIDI Parameter Change table (Normal Voice Element Data)

\$F0, \$43, \$1n, \$34, \$03, T2, \$00, N2, \$00, V2, \$F7

Note) n ; Device Number

V2 ; parameter value

T2 ; 0, e1, e0, 0, 0, 0, 0, 0

e1	e0	
0	0	Element 1
0	1	Element 2
1	0	Element 3
1	1	Element 4

	N2	data name	data range	note
0	00	ELVL	0~127	Element Level
1	01	ELDT	-7~+7 (s/m)	Element Detune
2	02	ELNS	-64~+63 (o/b)	Element Note Shift
3	03	ENLL	0~127 (note #)	—Element Note Limit—
4	04	ENLH	0~127 (note #)	Low Limit High Limit
5	05	EVLL	0~127 (note #)	—Element Velocity Limit—
6	06	EVLH	0~127 (note #)	Low Limit High Limit
7	07	PANNM	0~31	PAN data set table select Note: Only valid when selected for Multi
8	08	MCTEN	b0 : off/on	Micro Tuning switch
		OUT0SEL	b1 : off/on	Output 1 select
		OUT1SEL	b2 : off/on	Output 2 select
9	09	EFLN1EL	b0 : off/on	effect send line 1 select
		EFLN2EL	b1 : off/on	effect send line 2 select
		EFLN3EL	b2 : off/on	effect send line 3 select
		EFLN4EL	b3 : off/on	effect send line 4 select
10	0A	EFSDLV	0~127	effect send level
11	0B	EFSDVL	-7~+7 (s/m)	effect send velocity sense
12	0C	EFSDSC	-7~+7 (s/m)	effect send level scale

Note) (s/m) ; sign magnitude

(o/b) ; offset binary

< Table 1-5 >

MIDI Parameter Change table (Drum_Set)

\$F0, \$43, \$1n, \$34, \$04, T2, \$00, N2, V1, V2, \$F7

Note) n ; channel number

T2 ; MIDI note number

N2 ; parameter number

V1 ; MSB of parameter value (for parameters other than WAV*,
V1 will be \$00)

V2 ; LSB of parameter value

	N2	data name	data range	note
0	00	ALTGRP	b6 : 0~1	Alternate group
		OUT1	b1 : 0~1	Output 2 select
		OUT0	b0 : 0~1	Output 1 select
1	01	WSRC	0;Pr#1 1;Card 2;—3;Int	Wave Source
2	02	WAV	4;Pr#2 0~max. 255	Wave Number (V1;MSB V2;LS7bits)
3	03	WVL	0~127	Wave Volume
4	04	WTN	-64~+63 (o/b)	Wave Tuning
5	05	WNS	-48~+36 (o/b)	Wave Note Shift
6	06	WPN	-31~+31 (o/b)	Static PAN
7	07	EFLN1C_1	b0 : off/on	effect send line 1 select
		EFLN2C_1	b1 : off/on	effect send line 2 select
		EFLN3C_1	b2 : off/on	effect send line 3 select
		EFLN4C_1	b3 : off/on	effect send line 4 select
8	08	EFSDLVC_1	0~127	effect send level
9	09	EFSDVLC_1	-7~+7 (s/m)	effect send velocity sense

Note) (o/b) ; offset binary

< Table 1-6 >

MIDI Parameter Change table (AFM Element Common)

\$F0, \$43, \$1n, \$34, \$05, T2, \$00, N2, \$00, V2, \$F7

Note) n ; Device Number
 N2 ; parameter number
 V2 ; parameter value

T2	
\$00	Element 1
\$20	Element 2
\$40	Element 3
\$60	Element 4

	N2	data name	data range	note
0	00	ALGNUM	0~44	(127 is free algorithm number. Algorithm only via MIDI)
1	01	FPR1	0~63	---Pitch EG---
2	02	FPR2	0~63	key_on Rate 1
3	03	FPR3	0~63	key_on Rate 2
4	04	FPRR1	0~63	key_on Rate 3
5	05	FPL0	-64~+63 (o/b)	key_off Rate 1
6	06	FPL1	-64~+63 (o/b)	key_on Level 0
7	07	FPL2	-64~+63 (o/b)	key_on Level 1
8	08	FPL3	-64~+63 (o/b)	key_on Level 2
9	09	FPLR1	-64~+63 (o/b)	key_on Level 3
10	0A	FPEGR	0~3	key_off Level 1
11	0B	FPRS	0~7	Range (0: 8oct, 1: 2oct, 2: 1oct, 3: 1/2oct)
12	0C	FVPSW	off/on	Rate Scaling Velocity Switch
13	0D	FLFSPD	0~99	---Main LFO---
14	0E	FLFDLY	0~99	Speed
15	0F	FLFPMD	0~127	Delay time
16	10	FLFAMD	0~127	Pitch Modulation Depth
17	11	FLFFMD	0~127	Amplitude Modulation Depth
18	12	FLFWAV	0~5	Filter Modulation Depth
19	13	FLINTP	0~99	Wave
20	14	reserve		Initial Phase
21	15	SLFWD	0~3	---Sub LFO---
22	16	SLFS	0~127	Wave
23	17	SLFDM	delay/decay	Speed
24	18	SLFDT	0~99	delay mode/decay mode
25	19	SLPMD	0~127	Delay time/decay time Pitch Modulation Depth

< Table 1-7 >

MIDI Parameter Change table (AFM Element)

\$F0, \$43, \$1n, \$34, T1, T2, \$00, N2, V1, V2, \$F7

Note) n ; Device Number
 N2 ; parameter number
 V1 ; MSB of parameter value
 V2 ; LSB of parameter value

T1		T2	
\$06	OP6	\$00	Element 1
\$16	OP5	\$20	Element 2
\$26	OP4	\$40	Element 3
\$36	OP3	\$60	Element 4
\$46	OP2		
\$56	OP1		

	N2	data name	data range	note
0	00	R1	0~63	EG key_on Rate 1
1	01	R2	0~63	EG key_on Rate 2
2	02	R3	0~63	EG key_on Rate 3
3	03	R4	0~63	EG key_on Rate 4
4	04	RR1	0~63	EG key_off Rate 1
5	05	RR2	0~63	EG key_off Rate 2
6	06	L1	0~63	EG key_on Level 1
7	07	L2	0~63	EG key_on Level 2
8	08	L3	0~63	EG key_on Level 3
9	09	L4	0~63	EG key_on Level 4
10	0A	RL1	0~63	EG key_off Level 1
11	0B	RL2	0~63	EG key_off Level 2
12	0C	SLP	0~3	EG Sustain Loop Point
13	0D	HT	0~63 (disp: 63~0)	EG key_on/Hold Time
14	0E	L0	0~63	EG key_on Level 0
15	0F	RS	-7~+7 (s/m)	EG Rate Scaling
16	10	FAMS	0~7	Amplitude Modulation Sens.
17	11	VSON	-7~+7 (s/m)	Velocity Sensitivity
18	12	reserve		
19	13	ALGSR0	V2 b3~0	oscillator input 0 Source
20	14	ALGSR1	V1 b0, V2 b6~4	oscillator input 1 Source
21	15	ALGDST	V2 b1, 0	oscillator output Destination
22	16	OASCR0	V2 b3, 2	Out_Accumulator input 0 Source
23	17	OASCR1	V2 b4	Out_Accumulator input 1 Source
24	18	SHIFT0	V2 b5~3	oscillator input 0 Shift value
25	19	SHIFT1	V2 b2~0	oscillator input 1 Shift value
26	1A	OCR	0~7	output level Correction
27	1B	FWAVE	0~15	Waveform of oscillator
28	1C	FMLPMS	V2 b4~2	M_LFO Pitch Modulation Sens.
29	1D	PES	V2 b1	Pitch EG Switch
30	1E	FPM	V2 b0	frequency Mode
31	1F	KOE	V1 b0	initial phase set Enable
32	20	PHASE	V2 b6~0	initial Phase of oscillator
33	21	FPD	-15~+15 (s/m)	Pitch Detune

	N2	data name	data range	note
27	1B	TL	0~127	out_level
28	1C	BP1	0~127 (note #)	out_level scaling Break Point
29	1D	BP2	0~127 (note #)	out_level scaling Break Point
30	1E	BP3	0~127 (note #)	out_level scaling Break Point
31	1F	BP4	0~127 (note #)	out_level scaling Break Point
32	20	EGOS1	-128~+127 (2bytes)	out_level offset (BP1)
33	21	EGOS2	-128~+127 (2bytes)	out_level offset (BP2)
34	22	EGOS3	-128~+127 (2bytes)	out_level offset (BP3)
35	23	EGOS4	-128~+127 (2bytes)	out_level offset (BP4)
36	24	RVS	off/on	Rate Velocity Switch
37	25	FPC		frequency Course
38	26	FPF		frequency Fine

< Table 1-8 >

MIDI Parameter Change table (AFM Element)

\$F0, \$43, \$1n, #34, \$07, T2, \$00, N2, V1, V2, \$F7

Note) n ; Device Number
 N2 ; parameter number
 V1 ; MSB of parameter value
 V2 ; LSB of parameter value
 T2 ; table at right

T2	
\$00	Element 1
\$20	Element 2
\$40	Element 3
\$60	Element 4

[AWM generator unit]

	N2	data name	data range	note
0	00	WSOURCE	0; Pre1, 1; Card ,2; AFM, 3; Int, 4; Pre2	AWM Wave Source
1	01	AWMWAVE	V1: MSB1bit V2: LS7bits	AWM Wave number
2	02	PPM	normal/fixed	frequency Mode
3	03	PNOTE	0~127	fixed mode note #
4	04	PPF	-64~+63	frequency Fine
5	05	PMLPMS	0~7	pitch modulation sensitivity
6	06	PPR1	0~63	---Pitch EG---
7	07	PPR2	0~63	key_on Rate 1
8	08	PPR3	0~63	key_on Rate 2
9	09	PPRR1	0~63	key_on Rate 3
10	0A	PPL0	-64 ~+63 (o/b)	key_off Rate 1
11	0B	PPL1	-64 ~+63 (o/b)	key_on Level 0
12	0C	PPL2	-64 ~+63 (o/b)	key_on Level 1
13	0D	PPL3	-64 ~+63 (o/b)	key_on Level 2
14	0E	PPRL1	-64 ~+63 (o/b)	key_on Level 3
15	0F	PPEGR	1~3	key_off Level 1
16	10	PPRS	-7~+7	Range (1: 2oct, 2: 1oct, 3: 1/2oct)
17	11	PVPSW	off/on	Rate Scaling Velocity Switch
18	12	PLFSPD	0~99	---multi LFO---
19	13	PLFDLY	0~99	Speed
20	14	PLFPMD	0~127	Delay time
21	15	PLFAMD	0~127	Pitch Modulation Depth
22	16	PLFFMD	0~127	Amplitude Modulation Depth
23	17	PLFWAV	0~5	Filter Modulation Depth
24	18	PLINTP	0~99	Wave
25	19	reserve		Initial Phase

[Amplitude EG data]

	N2	data name	data range	note
25	4F	PAEGMD	normal/hold	EG mode
26	50	PAR1	0~63	key_on Rate 1 (attack/hold)
27	51	PAR2	0~63	key_on Rate 2 (decay)
28	52	PAR3	0~63	key_on Rate 3
29	53	PAR4	0~63	key_on Rate 4 (decay)
30	54	PARR1	0~63	key_off Rate 1 (release)
31	55	PAL2	0~63	key_on Level 2 (decay)
32	56	PAL3	0~63	key_on Level 3 (decay)
33	57	PARS	-7~+7	rate scaling
34	58	PABP1	0~127 (note #)	out_level scaling Break Point
35	59	PABP2	0~127 (note #)	out_level scaling Break Point
36	5A	PABP3	0~127 (note #)	out_level scaling Break Point
37	5B	PABP4	0~127 (note #)	out_level scaling Break Point
38	5C	PAOS21	-128~+127 (2bytes)	out_level scaling offset
39	5D	PAOS22	-128~+127 (2bytes)	out_level scaling offset
40	5E	PAOS23	-128~+127 (2bytes)	out_level scaling offset
41	5F	PAOS24	-128~+127 (2bytes)	out_level scaling offset
42	60	PAVSON	-7~+7 (s/m)	Velocity Sensitivity
43	61	PARVSW	off/on	Attack Rate Velocity Switch
44	62	PAMS	-7~+7 (s/m)	amplitude modulation sens.

< Table 1-9 >

MIDI Parameter Change table (Waveform)

\$F0, \$43, \$1n, \$34, \$0E, \$T2, 05, N2, 00, V2, \$F7

Note) n ; Device Number
 T2 ; Waveform number
 N2 ; parameter number
 V2 ; parameter value

	N2	data name	data range	note
0	00	WVNAME0	ascii	---Waveform Name--- .
1	01	WVNAME1	ascii	
2	02	WVNAME2	ascii	
3	03	WVNAME3	ascii	
4	04	WVNAME4	ascii	
5	05	WVNAME5	ascii	
6	06	WVNAME6	ascii	
7	07	WVNAME7	ascii	
8	08	WVFROM	—, 1~99	Sample From
9	09	WVTO	—, 1~99	Sample To

< Table 1-10 >

MIDI Parameter Change table (Effect Data)

\$F0, \$43, \$1n, \$34, \$08, \$00, \$00, N2, \$00, V2, \$F7

Note) n ; Device Number
 V2 ; parameter value

	N2	data name	data range	note
32	20	EFMODE	0~2	---Effect--- mode select (off, seri, palla) effect 1 type effect 1 parameter 1 effect 1 parameter 2 effect 1 parameter 3 effect 1 parameter 4 effect 1 parameter 5 effect 1 parameter 6 effect 1 parameter 7 effect 1 parameter 8 effect 1 parameter 9 effect 1 parameter 10 effect 1 output level 1 effect 1 output level 2 effect 2 type effect 2 parameter 1 effect 2 parameter 2 effect 2 parameter 3 effect 2 parameter 4 effect 2 parameter 5 effect 2 parameter 6 effect 2 parameter 7 effect 2 parameter 8 effect 2 parameter 9 effect 2 parameter 10 effect 2 mix level effect 2 output level 1 effect 2 output level 2 output 1 effect balance wet:dry output 2 effect balance wet:dry controller 1 parameter select controller 1 device assign controller 1 MIN controller 1 MAX controller 2 parameter select controller 2 device assign controller 2 MIN controller 2 MAX effect LFO wave select effect LFO speed effect LFO delay time effect LFO initial phase
33	21	EF1TYPE	0~60	
34	22	EF1PRM1		
36	24	EF1PRM2		
38	26	EF1PRM3		
40	28	EF1PRM4		
42	2A	EF1PRM5		
44	2C	EF1PRM6		
46	2E	EF1PRM7		
48	30	EF1PRM8		
50	32	EF1PRM9		
52	34	EF1PRM10		
54	36	EF1OUTLV1	0~100	
55	37	EF1OUTLV2	0~100	
56	38	EF2TYPE	0~60	
57	39	EF2PRM1		
59	3B	EF2PRM2		
61	3D	EF2PRM3		
63	3F	EF2PRM4		
65	41	EF2PRM5		
67	43	EF2PRM6		
69	45	EF2PRM7		
71	47	EF2PRM8		
73	49	EF2PRM9		
75	4B	EF2PRM10		
77	4D	EF2FBAL1	0~100	
78	4E	EF2OUTLV1	0~100	
79	4F	EF2OUTLV2	0~100	
80	50	OUT1EFBAL	0~100	
81	51	OUT2EFBAL	0~100	
82	52	CTRL1PRM	0~32	
83	53	CTRL1ASN	0~120, AT, VL, SC, LFO	
84	54	CTRL1MIN	0~99	
85	55	CTRL1MAX	0~99	
86	56	CTRL2PRM	0~32	
87	57	CTRL2ASN	0~120, AT, VL, SC, LFO	
88	58	CTRL2MIN	0~99	
89	59	CTRL2MAX	0~99	
90	5A	EFLFOWV	tri, dwn, up, squ, sin, S/H	
91	5B	EFLFOSP	0~99	
92	5C	EFLFODL	0~99	
93	5D	EFLFOPH	0~99, free	

< Table 1-11 >

MIDI Parameter Change table (Filter Data)

\$F0, \$43, \$1n, \$34, \$09, T2, \$00, N2, V1, V2, \$F7

Note) n ; Device Number
 N2 ; parameter number
 V1 ; MSB of parameter value
 V2 ; LSB of parameter value
 T2 ; O, E, E, O, O, N, N, N
 (E, N table at right)

E	N
00	Element 1
01	Element 2
10	Element 3
11	Element 4

000	AFM filt. 1
001	AFM filt. 2
010	AMF common
011	AWM filt. 1
100	AWM filt. 2
101	AWM common

[filter 1 & 2]

	N2	data name	data range	note
0	00	FTYPE	thru/LPF/HPF	filter type
1	01	FCTOF	0~127	cut_off frequency
2	02	FMODE	EG, LFO, EG-VA	filter mode
3	03	FR1	0~63	key_on Rate 1
4	04	FR2	0~63	key_on Rate 2
5	05	FR3	0~63	key_on Rate 3
6	06	FR4	0~63	key_on Rate 4
7	07	FRR1	0~63	key_off Rate 1
8	08	FRR2	0~63	key_off Rate 2
9	09	FL0	-64~+63 (o/b)	key_on cut_off Level 0
10	0A	FL1	-64~+63 (o/b)	key_on cut_off Level 1
11	0B	FL2	-64~+63 (o/b)	key_on cut_off Level 2
12	0C	FL3	-64~+63 (o/b)	key_off cut_off Level 3
13	0D	FL4	-64~+63 (o/b)	key_off cut_off Level 4
14	0E	FRL1	-64~+63 (o/b)	key_on cut_off Level 1
15	0F	FRL2	-64~+63 (o/b)	key_on cut_off Level 2
16	10	FRS	-7~+7	rate scaling
17	11	FBP1	0~127 (note #)	c_off_lv1 scaling Break Point
18	12	FBP2	0~127 (note #)	c_off_lv1 scaling Break Point
19	13	FBP3	0~127 (note #)	c_off_lv1 scaling Break Point
20	14	FBP4	0~127 (note #)	c_off_lv1 scaling Break Point
21	15	FOS1	-128~+127 (2bytes)	c_off_lv1 scaling offset
22	16	FOS2	-128~+127 (2bytes)	c_off_lv1 scaling offset
23	17	FOS3	-128~+127 (2bytes)	c_off_lv1 scaling offset
24	18	FOS4	-128~+127 (2bytes)	c_off_lv1 scaling offset

[filter common]

	N2	data name	data range	note
25	32	FRES	0~99	resonance
26	33	FVSON	-7~+7 (s/m)	Velocity Sensitivity
27	34	FCMS	-7~+7 (s/m)	Cut_off modulation sensitivity

< Table 1-12 >

MIDI Parameter Change table (Pan Data)

\$F0, \$43, \$1n, \$34, \$0A, T2, \$00, N2, \$00, V2, \$F7

Note) n ; Device Number
 T2 ; Memory number
 N2 ; parameter number
 V2 ; parameter value

	N2	data name	data range	note
0	00	PNSCSEL	velocity, note #, LFO	PAN source select
1	01	PNSCDPT	0~99	PAN source depth
2	02	PNDT	0~63	---EG--- key_on/Hold Time key_on Rate 1 key_on Rate 2 key_on Rate 3 key_on Rate 4 key_off Rate 1 key_off Rate 2 key_on Level 0 key_on Level 1 key_on Level 2 key_on Level 3 key_on Level 4 key_off Level 1 key_off Level 2 repeat segment
3	03	PNR1	0~63	
4	04	PNR2	0~63	
5	05	PNR3	0~63	
6	06	PNR4	0~63	
7	07	PNRR1	0~63	
8	08	PNRR2	0~63	
9	09	PNL0	-32~+31 (o/b)	
10	0A	PNL1	-32~+31 (o/b)	
11	0B	PNL2	-32~+31 (o/b)	
12	0C	PNL3	-32~+31 (o/b)	
13	0D	PNL4	-32~+31 (o/b)	
14	0E	PNRL1	-32~+31 (o/b)	
15	0F	PNRL2	-32~+31 (o/b)	
16	10	PNSLP	0~3	
17	11	PNNAM0	ascii	---Dynamic PAN Name--- .
18	12	PNNAM1	ascii	
19	13	PNNAM2	ascii	
20	14	PNNAM3	ascii	
21	15	PNNAM4	ascii	
22	16	PNNAM5	ascii	
23	17	PNNAM6	ascii	
24	18	PNNAM7	ascii	
25	19	PNNAM8	ascii	
26	1A	PNNAM9	ascii	

Note) (o/b) ; offset binary (invert the sign_bit of the 2's complement)

< Table 1-13 >

MIDI Parameter Change table (Micro Tuning Data)

\$F0, \$43, \$1n, \$34, \$0B, T2, N1, N2, V1, V2, \$F7

Note) n ; Device Number
 V1 ; MS7bits of parameter value
 T2 ; memory number
 V2 ; LS7bits of parameter value

	N1	N2	data name	data range	note
0	00	00	MCTC_2	0~10794	C_2
1	00	02	MCTC#_2	0~10794	C#_2
2	00	04	MCTD_2	0~10794	D_2
3	00	06	MCTD#_2	0~10794	D#_2
4	00	08	MCTE_2	0~10794	E_2
5	00	0A	MCTF_2	0~10794	F_2
6	00	0C	MCTF#_2	0~10794	F#_2
7	00	0E	MCTG_2	0~10794	G_2
8	00	10	MCTG#_2	0~10794	G#_2
9	00	12	MCTA_2	0~10794	A_2
10	00	14	MCTA#_2	0~10794	A#_2
11	00	16	MCTB_2	0~10794	B_2
12	00	18	MCTC_1	0~10794	C_1
13	00	1A	MCTC#_1	0~10794	C#_1
14	00	1C	MCTD_1	0~10794	D_1
15	00	1E	MCTD#_1	0~10794	D#_1
16	00	20	MCTE_1	0~10794	E_1
17	00	22	MCTF_1	0~10794	F_1
18	00	24	MCTF#_1	0~10794	F#_1
19	00	26	MCTG_1	0~10794	G_1
20	00	28	MCTG#_1	0~10794	G#_1
21	00	2A	MCTA_1	0~10794	A_1
22	00	2C	MCTA#_1	0~10794	A#_1
23	00	2E	MCTB_1	0~10794	B_1
24	00	30	MCTC_0	0~10794	C_0
25	00	32	MCTC#0	0~10794	C#0
26	00	34	MCTD_0	0~10794	D_0
27	00	36	MCTD#0	0~10794	D#0
28	00	38	MCTE_0	0~10794	E_0
29	00	3A	MCTF_0	0~10794	F_0
30	00	3C	MCTF#0	0~10794	F#0
31	00	3E	MCTG_0	0~10794	G_0
32	00	40	MCTG#0	0~10794	G#0
33	00	42	MCTA_0	0~10794	A_0
34	00	44	MCTA#0	0~10794	A#0
35	00	46	MCTB_0	0~10794	B_0
36	00	48	MCTC_1	0~10794	C_1
37	00	4A	MCTC#1	0~10794	C#1
38	00	4C	MCTD_1	0~10794	D_1
39	00	4E	MCTD#1	0~10794	D#1
40	00	50	MCTE_1	0~10794	E_1
41	00	52	MCTF_1	0~10794	F_1
42	00	54	MCTF#1	0~10794	F#1
43	00	56	MCTG_1	0~10794	G_1
44	00	58	MCTG#1	0~10794	G#1
45	00	5A	MCTA_1	0~10794	A_1
46	00	5C	MCTA#1	0~10794	A#1
47	00	5E	MCTB_1	0~10794	B_1
48	00	60	MCTC_2	0~10794	C_2
49	00	62	MCTC#2	0~10794	C#2
50	00	64	MCTD_2	0~10794	D_2
51	00	66	MCTD#2	0~10794	D#2
52	00	68	MCTE_2	0~10794	E_2
53	00	6A	MCTF_2	0~10794	F_2
54	00	6C	MCTF#2	0~10794	F#2
55	00	6E	MCTG_2	0~10794	G_2
56	00	70	MCTG#2	0~10794	G#2
57	00	72	MCTA_2	0~10794	A_2
58	00	74	MCTA#2	0~10794	A#2
59	00	76	MCTB_2	0~10794	B_2
60	00	78	MCTC_3	0~10794	C_3
61	00	7A	MCTC#3	0~10794	C#3
62	00	7C	MCTD_3	0~10794	D_3
63	00	7E	MCTD#3	0~10794	D#3
64	01	00	MCTE_3	0~10794	E_3
65	01	02	MCTF_3	0~10794	F_3
66	01	04	MCTF#3	0~10794	F#3
67	01	06	MCTG_3	0~10794	G_3
68	01	08	MCTG#3	0~10794	G#3
69	01	0A	MCTA_3	0~10794	A_3
70	01	0C	MCTA#3	0~10794	A#3
71	01	0E	MCTB_3	0~10794	B_3
72	01	10	MCTC_4	0~10794	C_4
73	01	12	MCTC#4	0~10794	C#4
74	01	14	MCTD_4	0~10794	D_4
75	01	16	MCTD#4	0~10794	D#4
76	01	18	MCTE_4	0~10794	E_4
77	01	1A	MCTF_4	0~10794	F_4
78	01	1C	MCTF#4	0~10794	F#4
79	01	1E	MCTG_4	0~10794	G_4
80	01	20	MCTG#4	0~10794	G#4
81	01	22	MCTA_4	0~10794	A_4
82	01	24	MCTA#4	0~10794	A#4
83	01	26	MCTB_4	0~10794	B_4
84	01	28	MCTC_5	0~10794	C_5
85	01	2A	MCTC#5	0~10794	C#5
86	01	2C	MCTD_5	0~10794	D_5
87	01	2E	MCTD#5	0~10794	D#5
88	01	30	MCTE_5	0~10794	E_5
89	01	32	MCTF_5	0~10794	F_5
90	01	34	MCTF#5	0~10794	F#5
91	01	36	MCTG_5	0~10794	G_5
92	01	38	MCTG#5	0~10794	G#5
93	01	3A	MCTA_5	0~10794	A_5
94	01	3C	MCTA#5	0~10794	A#5
95	01	3E	MCTB_5	0~10794	B_5
96	01	40	MCTC_6	0~10794	C_6

	N1	N2	data name	data range	note
97	01	42	MCTC#6	0~10794	C#6
98	01	44	MCTD_6	0~10794	D_6
99	01	46	MCTD#6	0~10794	D#6
100	01	48	MCTE_6	0~10794	E_6
101	01	4A	MCTF_6	0~10794	F_6
102	01	4C	MCTF#6	0~10794	F#6
103	01	4E	MCTG_6	0~10794	G_6
104	01	50	MCTG#6	0~10794	G#6
105	01	52	MCTA_6	0~10794	A_6
106	01	54	MCTA#6	0~10794	A#6
107	01	56	MCTB_6	0~10794	B_6
108	01	58	MCTC_7	0~10794	C_7
109	01	5A	MCTC#7	0~10794	C#7
110	01	5C	MCTD_7	0~10794	D_7
111	01	5E	MCTD#7	0~10794	D#7
112	01	60	MCTE_7	0~10794	E_7
113	01	62	MCTF_7	0~10794	F_7
114	01	64	MCTF#7	0~10794	F#7
115	01	66	MCTG_7	0~10794	G_7
116	01	68	MCTG#7	0~10794	G#7
117	01	6A	MCTA_7	0~10794	A_7
118	01	6C	MCTA#7	0~10794	A#7
119	01	6E	MCTB_7	0~10794	B_7
120	01	70	MCTC_8	0~10794	C_8
121	01	72	MCTC#8	0~10794	C#8
122	01	74	MCTD_8	0~10794	D_8
123	01	76	MCTD#8	0~10794	D#8
124	01	78	MCTE_8	0~10794	E_8
125	01	7A	MCTF_8	0~10794	F_8
126	01	7C	MCTF#8	0~10794	F#8
127	01	7E	MCTG_8	0~10794	G_8
128	02	00	MTNAM0	ascii	—Micro Tuning Name—
129	02	01	MTNAM1	ascii	"
130	02	02	MTNAM2	ascii	"
131	02	03	MTNAM3	ascii	"
132	02	04	MTNAM4	ascii	"
133	02	05	MTNAM5	ascii	"
134	02	06	MTNAM6	ascii	"
135	02	07	MTNAM7	ascii	"
136	02	08	MTNAM8	ascii	"
137	02	09	MTNAM9	ascii	"

< Table 1-14 >

MIDI Parameter Change table (Switch Remote)

\$F0, \$43, \$1n, \$34, \$0D, \$00, \$00, N2, \$00, V2, \$F7

Note) n ; Device Number
 N2 ; parameter number
 V2 ; parameter value data range : off (\$00~\$3F), on (\$40~\$7F)

	N2	sw.#	note		N2	sw.#	note
0	00	PSW 1	VOICE	36	24	PSW37	ENTER
1	01	PSW 2	MULTI	37	25	PSW38	1
2	02	PSW 3	SONG	38	26	PSW39	2
3	03	PSW 4	PATTERN	39	27	PSW40	3
4	04	PSW 5	UTILITY	40	28	PSW41	4
5	05	PSW 6	EDIT	41	29	PSW42	5
6	06	PSW 7	STORE	42	2A	PSW43	6
7	07	PSW 8	EFFECT	43	2B	PSW44	7
8	08	PSW 9	I <	44	2C	PSW45	8
9	09	PSW10	<<	45	2D	PSW46	9
10	0A	PSW11	LOCATE	46	2E	PSW47	INTERNAL
11	0B	PSW12	>>	47	2F	PSW48	CARD
12	0C	PSW13	RECORD	48	30	PSW49	PRESET 1
13	0D	PSW14	STOP	49	31	PSW50	PRESET 2
14	0E	PSW15	RUN	50	32	PSW51	A
15	0F	PSW16	SHIFT	51	33	PSW52	B
16	10	PSW17	F1	52	34	PSW53	C
17	11	PSW18	F2	53	35	PSW54	D
18	12	PSW19	F3	54	36	PSW55	1
19	13	PSW20	F4	55	37	PSW56	2
20	14	PSW21	F5	56	38	PSW57	3
21	15	PSW22	F6	57	39	PSW58	4
22	16	PSW23	F7	58	3A	PSW59	5
23	17	PSW24	F8	59	3B	PSW60	6
24	18	PSW25	EXIT	60	3C	PSW61	7
25	19	PSW26	PAGE <	61	3D	PSW62	8
26	1A	PSW27	PAGE >	62	3E	PSW63	9
27	1B	PSW28	JUMP	63	3F	PSW64	10
28	1C	PSW29	-1	64	40	PSW65	11
29	1D	PSW30	↑	65	41	PSW66	12
30	1E	PSW31	+1	66	42	PSW67	13
31	1F	PSW32	↑	67	43	PSW68	14
32	20	PSW33	↓	68	44	PSW69	15
33	21	PSW34	↓	69	45	PSW70	16
34	22	PSW35	0	70	46	PSW71	DIAL DEC
35	23	PSW36	±	71	47	PSW72	DIAL INC

< Table 1-15 >

MIDI Parameter Change table (Master control)

\$F0, \$43, \$1n, \$34, \$0E, T2, N1, N2, V1, V2, \$F7

Note) n ; channel number
 T2 ; control number
 N1 ; parameter type
 N2 ; parameter number
 V1 ; MS7bits of parameter value
 V2 ; LS7bits of parameter value

[Filter section] T2=00

	N1	N2	data name	data range	note
0	00	00	FILPGMMS	0-b'11111111	Program Change Filter ch 16-9
1	00	01	FILPGMLS	0-b'11111111	0:off 1:on ch 8-1
2	00	02	FILCONMS	0-b'11111111	Control Change Filter ch 16-9
3	00	03	FILCONLS	0-b'11111111	0:off 1:onch 8-1
4	00	04	FILPNDMS	0-b'11111111	Pitch Bend Filter ch 16-9
5	00	05	FILPNDLS	0-b'11111111	0:off 1:onch 8-1
6	00	06	FILSUSMS	0-b'11111111	Sustain Pedal Filter ch 16-9
7	00	07	FILSUSLS	0-b'11111111	0:off 1:onch 8-1
8	00	08	FILATCHMS	0-b'11111111	After Touch Filter ch 16-9
9	00	09	FILATCHLS	0-b'11111111	0:off 1:onch 8-1
10	00	0A	FILMVOLMS	0-b'11111111	Main Volume Filter ch 16-9
11	00	0B	FILMVOLLS	0-b'11111111	0:off 1:onch 8-1

[controller element section]

T2=0, 0, n, n, 0, t, t, t

n	
00	MIDI-1
01	MIDI-2
10	MIDI-3
11	MIDI-4

t	
000	Control 1
001	Control 2
010	Control 3
011	Control 4
100	Control 5
101	Control 6
110	Control 7
111	Control 8

	N1	N2	data name	data range	note
0	02	00	CONENABL	0-B'00011111	Control Enable 0:off 1:on Bit 0 = midi off/on 1 = pc off/on 2 = Ms/Ls off/on 3 = volume off/on 4 = MDR off/on
1	02	01	MIDITCH	0-15	MIDI Transmit Channel
2	02	02	NLIML	0-127	Note Limit Low
3	02	03	NLIMH	0-127	Note Limit High
4	02	04	VLIML	1-127	Velocity Limit Low
5	02	05	VLIMH	1-127	Velocity Limit High
6	02	06	XPOSE	0-127	Transpose [center 64 (40h)]
7	02	07	VELCURV	0-3	Velocity Curve
8	02	08	AFTCURV	0-3	After Touch Curve
9	02	09	MAINVOL	0-127	MAIN Volume
10	02	0A	BANK	0-16383	Bank Select
11	02	0C	PCNUM	0-127	Program Change Number
12	02	0D	MDRNUM	0-98	MDR Number

[Control name section]

T2=0, 0, 0, 0, 0, t, t, t

t	
000	Control 1
001	Control 2
010	Control 3
011	Control 4
100	Control 5
101	Control 6
110	Control 7
111	Control 8

	N1	N2	data name	data range	note
0	03	00	CNAM0	ascii	Control Name
1	03	01	CNAM1	ascii	
2	03	02	CNAM2	ascii	
3	03	03	CNAM3	ascii	
4	03	04	CNAM4	ascii	
5	03	05	CNAM5	ascii	
6	03	06	CNAM6	ascii	
7	03	07	CNAM7	ascii	
8	03	08	CNAM8	ascii	
9	03	09	CNAM9	ascii	

[Control voice section]

T2=0, 0, 0, 0, 0, t, t, t

t	
000	Control 1
001	Control 2
010	Control 3
011	Control 4
100	Control 5
101	Control 6
110	Control 7
111	Control 8

	N1	N2	data name	data range	note
0	04	00	PTM	off/on	Program
		01	VM	voice/multi	Program Mode
		02	VMEM	I/C/P1/P2	Voice Memory
		03	VNUM	0-63	Voice Number
		04	MMEM	I/C/P	Multi Memory
		05	MNUM	0-15	Multi Number

< Table 1-16 >

MIDI Parameter Change table (System Setup)

\$F0, \$43, \$1n, \$34, \$0F, \$00, \$00, N2, \$00, V2, \$F7

Note) n ; channel number
 N2 ; parameter number
 V2 ; parameter value

Note) Exception ; Master_Tuning (at transmit) \$F0 \$43, \$1N, \$04, DT, \$F7 (DX1 Master Tuning is used)

	N2	data name	data range	note
0	00	GRTMSU 0	ascii	Upper * * * * *
1	01	GRTMSU 1	ascii	Upper * * * * *
2	02	GRTMSU 2	ascii	Upper * * * * *
3	03	GRTMSU 3	ascii	Upper * * * * *
4	04	GRTMSU 4	ascii	Upper * * * * *
5	05	GRTMSU 5	ascii	Upper * * * * *
6	06	GRTMSU 6	ascii	Upper * * * * *
7	07	GRTMSU 7	ascii	Upper * * * * *
8	08	GRTMSU 8	ascii	Upper * * * * *
9	09	GRTMSU 9	ascii	Upper * * * * *
10	0A	GRTMSU10	ascii	Upper * * * * *
11	0B	GRTMSU11	ascii	Upper * * * * *
12	0C	GRTMSU12	ascii	Upper * * * * *
13	0D	GRTMSU13	ascii	Upper * * * * *
14	0E	GRTMSU14	ascii	Upper * * * * *
15	0F	GRTMSU15	ascii	Upper * * * * *
16	10	GRTMSU16	ascii	Upper * * * * *
17	11	GRTMSU17	ascii	Upper * * * * *
18	12	GRTMSU18	ascii	Upper * * * * *
19	13	GRTMSU19	ascii	Upper * * * * *
20	14	GRTMSL 0	ascii	Lower * * * * *
21	15	GRTMSL 1	ascii	Lower * * * * *
22	16	GRTMSL 2	ascii	Lower * * * * *
23	17	GRTMSL 3	ascii	Lower * * * * *
24	18	GRTMSL 4	ascii	Lower * * * * *
25	19	GRTMSL 5	ascii	Lower * * * * *
26	1A	GRTMSL 6	ascii	Lower * * * * *
27	1B	GRTMSL 7	ascii	Lower * * * * *
28	1C	GRTMSL 8	ascii	Lower * * * * *
29	1D	GRTMSL 9	ascii	Lower * * * * *
30	1E	GRTMSL10	ascii	Lower * * * * *
31	1F	GRTMSL11	ascii	Lower * * * * *
32	20	GRTMSL12	ascii	Lower * * * * *
33	21	GRTMSL13	ascii	Lower * * * * *
34	22	GRTMSL14	ascii	Lower * * * * *
35	23	GRTMSL15	ascii	Lower * * * * *
36	24	GRTMSL16	ascii	Lower * * * * *
37	25	GRTMSL17	ascii	Lower * * * * *
38	26	GRTMSL18	ascii	Lower * * * * *
39	27	GRTMSL19	ascii	Lower * * * * *
40	28	MNSFT	-64-63 (o/b)	Master Note Shift
41	29	MTUNE	-64-63 (o/b)	Master Fine Tuning (receive only)
42	2A	FIXVEL	off, 1-127	Fixed velocity
43	2B	VELCRV	0-7	Velocity Curve select
44	2C	MW2MCN	0-120	MIDI Control Number Assign
45	2D	FSASN	0-120	Modulation Wheel 2 Foot Switch assign
46	2E	EDCONFWS	off/on	Edit Confirm switch
47	2F	TXCH	0-15	keyboard transmit channel
48	30	VRCH	0-15, omni	Voice Receive channel
49	31	LOCAL	off/on	Local switch
50	32	DVCNUM	off, 0-15, all	Device number
51	33	NTSW	all/odd/even	Note_Even_Odd switch
52	34	BLKMPRT	off/on	Bulk data Memory protect switch
53	35	PGCMOD	off/on	Program Change mode
54	36	reserve		
55	37	PATPRT	off/on	Bulk Data Pattern Protect Switch

	N2	data name	data range	note
56	38	CONTHOLD	off/on	Controller Hold
57	39	MEMALOC	0~127	memory allocate for MDR (x4Kbyte)
58 59 60 61 62 63	3A 3B 3C 3D 3E 3F	reserve reserve reserve reserve reserve reserve		
	40	EBYP	0~2	Effect Bypass Sw (param cng only) 0: off 1: on Normal 2: on OUT-1 Direct OUT-2 Effect

Note) (o/b) ; offset binary (invert the sign_bit of the 2's complement)

< Table 1-17 >

MIDI Parameter Change table (Sample)

\$F0, \$43, \$1n, \$34, \$0E, \$T2, 06, N2, V1, V2, \$F7

Note) n ; Device Number
T2 ; Sample number
N2 ; parameter number
V1 ; MS7bits of parameter value
V2 ; LS7bits of parameter value

	N2	data name	data range	note
0 1 2	00 02 04	HIKEY ORIKEY PITCH	1~127 (note #) 0~127 (note #) -5376~+5376 (2bytes, o/b)	High key Original key Pitch
3	08	LOOPMODE	b2 ; 0 forward 1 backward b1 ; 0 loop off 1 loop on b0 ; 0 normal 1 alternate	Loop type, Mode
4 5	12 1C	VOL LOWKEY	0~127 1~127 (note #)	Volume Low key
6 7 8 9 10 11 12 13	1E 1F 20 21 22 23 24 25	SANAM0 SANAM1 SANAM2 SANAM3 SANAM4 SANAM5 SANAM6 SANAM7	ascii ascii ascii ascii ascii ascii ascii ascii	—Sample Name— *

< Table 2-1 >

Voice Bulk Dump

Note) Memory_type internal ; \$00
preset 1 ; \$02
preset 2 ; \$03
Edit Buffer ; \$7F (Used only when transmitting from SY99. Memory# is transmitted as \$00, ignored when receiving.)

When receiving Bulk dump, if Memory_type is other than \$7F, this is processed as Internal.

Note) Memory# \$00~\$0F ; Bank A, 1~16
\$10~\$1F ; Bank B, 1~16
\$20~\$2F ; Bank C, 1~16
\$30~\$3F ; Bank D, 1~16

Note) When receiving Bulk dump, bit 6 of Memory# is ignored.

(1) 1AFM

	data	Element1 data	Element1 data
0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33 34 35 36 37 38 39 40 41 42 43 44 45 46 47 48 49 50 51 52 53 54 55 56 57 58 59 60 61 62 63 64 65 66 67 68 69 70 71 72 73 74 75 76 77 78 79 80 81 82 83 84 85 86 87 88 89 90 91 92 93 94 95 96 97 98 99 100 101 102 103 104 105 106 107 108 109 110 111 112 113 114 115 116 117 118 119 120 121 122 123 124 125 126 127 128 129 130 131 132 133 134 135 136 137 138 139 140 141 142 143 144 145 146 147 148 149 150 151 152 153 154 155 156 157 158 159 160 161 162 163 164 165 166 167 168 169 170 171 172 173 174 175 176 177 178 179 180 181 182 183 184 185 186 187 188 189 190 191 192 193 194 195 196 197 198 199 200 201 202 203 204 205 206 207 208 209 210 211 212 213 214 215 216 217 218 219 220 221 222 223 224 225 226 227 228 229 230 231 232 233 234 235 236 237 238 239 240 241 242 243 244 245 246 247 248 249 250 251 252 253 254 255 256 257 258 259 260 261 262 263 264 265 266 267 268 269 270 271 272 273 274 275 276 277 278 279 280 281 282 283 284 285 286 287 288 289 290 291 292 293 294 295 296 297 298 299 300 301 302 303 304 305 306 307 308 309 310 311 312 313 314 315 316 317 318 319 320 321 322 323 324 325 326 327 328 329 330 331 332 333 334 335 336 337 338 339 340 341 342 343 344 345 346 347 348 349 350 351 352 353 354 355 356 357 358 359 360 361 362 363 364 365 366 367 368 369 370 371 372 373 374 375 376 377 378 379 380 381 382 383 384 385 386 387 388 389 390 391 392 393 394 395 396 397 398 399 400 401 402 403 404 405 406 407 408 409 410 411 412 413 414 415 416 417 418 419 420 421 422 423 424 425 426 427 428 429 430 431 432 433 434 435 436 437 438 439 440 441 442 443 444 445 446 447 448 449 450 451 452 453 454 455 456 457 458 459 460 461 462 463 464 465	\$F0 \$43 \$0N \$7A byte count L M V C Memory_type Memory# Refer to 00~0A of Table 1-3. \$00 or \$03 (ELMODE) VNAMO Refer to Dummy of Table 1-10. EFMODE Refer to 28~3F of Table 1-3. WPBR VVOL AWMID high 7bit AWMID low 7bit Refer to 00~08 of Table 1-4. ELVLO MCTENO	Refer to 00~26 of Table 1-7. OP6_R1 reserve OP6_ALGSRG (MSB) OP6_ALGSRG (low 7bits) OP6_ALGDST OP6_FMPMS OP6_KOE OP6_PHASE OP6_FPD OP6_BP4 OP6_EGOS1 (MSB) OP6_EGOS1 (low 7bits) OP6_EGOS2 (MSB) OP6_EGOS2 (low 7bits) OP6_EGOS3 (MSB) OP6_EGOS3 (low 7bits) OP6_EGOS4 (MSB) OP6_EGOS4 (low 7bits) OP6_RVSW OP6_FPC OP6_PPF Refer to 00~26 of Table 1-7. OP5_R1 Refer to 00~26 of Table 1-7. OP4_R1 Refer to 00~26 of Table 1-7. OP3_R1 Refer to 00~26 of Table 1-7. OP2_R1 Refer to 00~26 of Table 1-7. OP1_R1	Refer to 00~19 of Table 1-6. ALGNUM Refer to 00~18 of Table 1-11. FL1_FTYPE FL1_FBP4 FL1_FOS1 (MSB) FL1_FOS1 (low 7bits) FL1_FOS2 (MSB) FL1_FOS2 (low 7bits) FL1_FOS3 (MSB) FL1_FOS3 (low 7bits) FL1_FOS4 (MSB) FL1_FOS4 (low 7bits) Refer to 00~18 of Table 1-11. FL2_FTYPE FL2_FOS4 (MSB) FL2_FOS4 (low 7bits) Refer to 32~34 of Table 1-11. FFRES FFVSON FFCMS check_sum \$F7

(2) 2AFM

data		Element1 data	
0	\$F0	116	Same as (1) AFM 107~463
1	\$43	↓	
2	\$0N	↓	
3	\$7A	472	
4	byte count		
5			
6	L	Element2 data	
7	M	473	Same as (1) AFM 107~463
8	-	↓	
9	-	829	
10	8		
11	1		
12	0	830	check_sum
13	1	831	\$F7
14	V		
15	C		
16	\$00		
29			
30	Memory_type		
31	Memory#		
32	\$01 or \$04 (ELMODE)		
33	Same as (1) AFM 33~97		
↓			
↓			
97			
98	Refer to 00~08 of Table 1-4. ELVL0		
↓			
106	MCTEN0		
107	Refer to 00~08 of Table 1-4. ELVL1		
↓			
115	MCTEN1		

(4) 1AWM

data		Element1 data	
0	\$F0	Refer to 00~19 of Table 1-8. WSOURCE PCMWAWE (MSB) PCMWAWE (low 7bits) PPM	
1	\$43		
2	\$0N		
3	\$7A		
4	byte count	Same as (1) AFM 403~463	
5			
6	L		
7	M		
8	-	134	Same as (1) AFM 403~463
9	-	↓	
10	8	194	
11	1		
12	0		
13	1		
14	V		
15	C		
16	\$00		
29			
30	Memory_type		
31	Memory#		
32	\$05 (ELMODE)		
33	Same as (1) AFM 33~97		
↓			
↓			
97			
98	Refer to 00~08 of Table 1-4. ELVL0		
↓			
106	MCTEN0		
195	Refer to 4F~62 of Table 1-8. PAEGMD		
207	PABP4		
208	PAOS21 (MSB)		
209	PAOS21 (low 7bits)		
210	PAOS22 (MSB)		
211	PAOS22 (low 7bits)		
212	PAOS23 (MSB)		
213	PAOS23 (low 7bits)		
214	PAOS24 (MSB)		
215	PAOS24 (low 7bits)		
216	PAVSON		
217	PARVSW		
218	PAMS		
219	check_sum		
220	\$F7		

(3) 4AFM

Note) Memory# ; \$30~\$3F

When receiving, cancel if Memory# is other than the above.

data		Element1 data	
0	\$F0	134	Same as (1) AFM 107~463
1	\$43	↓	
2	\$0N	↓	
3	\$7A	490	
4	byte count	Element2 data	
5		491	Same as (1) AFM 107~463
6		↓	
7		847	
8			
9	-	Element3 data	
10	8	848	Same as (1) AFM 107~463
11	1	↓	
12	0	1204	
13	1		
14	V	Element4 data	
15	C	1205	Same as (1) AFM 107~463
16	\$00	↓	
29		1561	
30	Memory_type	1562	
31	Memory#	1563	
32	\$02 (ELMODE)		
33	Same as (1) AFM 33~97		
↓			
↓			
97			
98	Refer to 00~08 of Table 1-4. ELVL0		
↓			
106	MCTEN0		
107	Refer to 00~08 of Table 1-4. ELVL1		
↓			
115	MCTEN1		
116	Refer to 00~08 of Table 1-4. ELVL2		
↓			
124	MCTEN2		
125	Refer to 00~08 of Table 1-4. ELVL3		
↓			
133	MCTEN3		

(5) 2AWM

data		Element1 data	
0	\$F0	116	Same as (1) 1AWM 107~218
1	\$43	↓	
2	\$0N	↓	
3	\$7A	227	
4	byte count	Element2 data	
5		Same as (1) 1AWM 107~218	
6	228		
7	↓		
8	339		
9			
10	8		
11	1		
12	0	check_sum \$F7	
13	1		
14	V		
15	C		
16	\$00		
29			
30	Memory_type		
31	Memory#		
32	\$06 (ELMODE)		
33	Same as (1) AFM 33~97		
↓			
97			
98	Refer to 00~08 of Table 1-4. ELVL0		
↓			
106	MCTEN0		
107	Refer to 00~08 of Table 1-4. ELVL1		
↓			
115	MCTEN1		

(6) 4AWM

data		Element1 data	
0	\$F0	134	Same as (1) 1AWM 107~218
1	\$43	↓	
2	\$0N	245	Element2 data
3	\$7A	↓	
4	byte count	246	Same as (1) 1AWM 107~218
5	L	↓	
6	M	357	Element3 data
7	-	↓	
8	-	358	Same as (1) 1AWM 107~218
9	-	↓	
10	8	469	Element4 data
11	1	↓	
12	0	470	Same as (1) 1AWM 107~218
13	1	↓	
14	V	581	check_sum \$F7
15	C	↓	
16	\$00	582	
17	Memory_type	↓	
18	Memory#	583	
32	\$07 (ELMODE)		
33	Same as (1) AFM 33~97		
↓			
97			
98	Refer to 00~08 of Table 1-4. ELVL0		
↓	MCTEN0		
106			
107	Refer to 00~08 of Table 1-4. ELVL1		
↓	MCTEN1		
115			
116	Refer to 00~08 of Table 1-4. ELVL2		
↓	MCTEN2		
124			
125	Refer to 00~08 of Table 1-4. ELVL3		
↓	MCTEN3		
133			

(7) 1AFM_1AWM

data		Element1 data	
0	\$F0	116	Same as (1) AFM 107~463
1	\$43	↓	
2	\$0N	472	Element2 data
3	\$7A	↓	
4	byte count	473	Same as (4) 1AWM 107~218
5	L	↓	
6	M	584	check_sum \$F7
7	-	↓	
8	-	585	
9	-	↓	
10	8	586	
11	1		
12	0		
13	1		
14	V		
15	C		
16	\$00		
17	Memory_type		
18	Memory#		
32	\$08 (ELMODE)		
33	Same as (1) AFM 33~97		
↓			
97			
98	Refer to 00~08 of Table 1-4. ELVL0		
↓	MCTEN0		
106			
107	Refer to 00~08 of Table 1-4. ELVL1		
↓	MCTEN1		
115			

(8) 2AFM_2AWM

Note) Memory# ; \$30~\$3F

When receiving, cancel if Memory# is other than the above.

data		Element1 data	
0	\$F0	134	Same as (1) AFM 107~463
1	\$43	↓	
2	\$0N	490	Element2 data
3	\$7A	↓	
4	byte count	491	Same as (1) AFM 107~463
5	L	↓	
6	M	847	Element3 data
7	-	↓	
8	-	848	Same as (1) 1AWM 107~218
9	-	↓	
10	8	959	Element4 data
11	1	↓	
12	0	960	Same as (1) 1AWM 107~218
13	1	↓	
14	V	1071	check_sum \$F7
15	C	↓	
16	\$00	1072	
17	Memory_type	↓	
18	Memory#	1073	
32	\$09 (ELMODE)		
33	Same as (1) AFM 33~97		
↓			
97			
98	Refer to 00~08 of Table 1-4. ELVL0		
↓	MCTEN0		
106			
107	Refer to 00~08 of Table 1-4. ELVL1		
↓	MCTEN1		
115			
116	Refer to 00~08 of Table 1-4. ELVL2		
↓	MCTEN2		
124			
125	Refer to 00~08 of Table 1-4. ELVL3		
↓	MCTEN3		
133			

(9) Drum_set

data		Drum_set data		Drum_set data	
0	\$F0	98	ALTC_1, OUT*C_1	194	ALTC_2, OUT*C_2
1	\$43	99	WSRCC_1	↓	↓
2	\$0N	100	WAVC_1 (MSB)	↓	↓
3	\$7A	101	WAVC_1 (low 7bits)	↓	↓
4	byte count	102	WVLC_1	290	ALTC_3, OUT*C_3
5	L	103	WTNC_1	↓	↓
6	M	104	WNSC_1	↓	↓
7	-	105	WPNC_1	↓	↓
8	-	106	ALTC#1, OUT*C#1	386	ALTC_4, OUT*C_4
9	-	↓	↓	↓	↓
10	8	114	ALTD_1, OUT*D_1	↓	↓
11	1	↓	↓	482	ALTC_5, OUT*C_5
12	0	122	ALTD#1, OUT*D1	↓	↓
13	1	↓	↓	↓	↓
14	V	130	ALTE_1, OUT*E_1	578	ALTC_6, OUT*C_6
15	C	↓	↓	↓	↓
16	\$00	138	ALTF_1, OUT*F_1	585	WPNC_6
17	Memory_type	↓	↓	586	check_sum \$F7
18	Memory#	146	ALTF#1, OUT*F#1	587	
32	\$0A (ELMODE)	↓	↓		
33	Same as (1) AFM 33~71	154	ALTG_1, OUT*G_1		
↓		↓	↓		
71		162	ALTG#1, OUT*G#1		
72	\$00 (ctrl)	↓	↓		
87	VLASN	170	ALTA_1, OUT*A_1		
88	VLLML	↓	↓		
89	↓	178	ALTA#1, OUT*A#1		
90	\$00 (norm)	↓	↓		
95	↓	186	ALTB_1, OUT*B_1		
96	AWMID high 7	↓	↓		
97	AWMID low 7	↓	↓		

(10) dump request

	data
0	\$F0
1	\$43
2	\$2N
3	\$7A
4	L
5	M
6	-
7	-
8	8
9	1
10	0
11	1
12	V
13	C
14	↓
15	↓
16	↓
17	↓
18	↓
19	↓
20	↓
21	↓
22	↓
23	↓
24	↓
25	↓
26	↓
27	↓
28	Memory type
29	Memory#
30	\$F7

* Memory type = not \$7F (edit buffer)

< Table 2-2 >

Additional Voice Bulk Dump

Note) Memory_type internal ; \$00
 preset 1 ; \$02
 preset 2 ; \$03
 Edit Buffer ; \$7F (Used only when transmitting from SY99. Memory# is transmitted as \$00, ignored when receiving.)

For WSOURCE, AWMWAV, data which has the same name exists in the Voice Bulk Dump which is the AWM Element data shown in table 1-8, but when Voice Bulk and Additional Voice Bulk are transmitted together, Additional bulk WSOURCE, AWMWAV becomes effective.

When receiving Bulk dump, if Memory_type is other than \$7F, this is processed as Internal.

Note) Memory# \$00~\$0F ; Bank A, 1~16
 \$10~\$1F ; Bank B, 1~16
 \$20~\$2F ; Bank C, 1~16
 \$30~\$3F ; Bank D, 1~16

Note) When receiving Bulk dump, bit 6 of Memory# is ignored.

(1) 1AFM & 1AWM

	data
0	\$F0
1	\$43
2	\$0N
3	\$7A
4	byte count
5	↓
6	L
7	M
8	-
9	-
10	0
11	0
12	4
13	0
14	V
15	C
16	↓
17	↓
18	↓
19	↓
20	↓
21	↓
22	↓
23	↓
24	↓
25	↓
26	↓
27	↓
28	Memory_type
29	Memory#
30	Memory#
31	Memory#
32	Refer to 00 of Table 1-3. \$00, \$03, \$05 (ELMODE)
33	Refer to 20~5D of Table 1-10. EFMODE
34	↓
35	↓
36	↓
37	↓
38	↓
39	↓
40	↓
41	↓
42	↓
43	↓
44	↓
45	↓
46	↓
47	↓
48	↓
49	↓
50	↓
51	↓
52	↓
53	↓
54	↓
55	↓
56	↓
57	↓
58	↓
59	↓
60	↓
61	↓
62	↓
63	↓
64	↓
65	↓
66	↓
67	↓
68	↓
69	↓
70	↓
71	↓
72	↓
73	↓
74	↓
75	↓
76	↓
77	↓
78	↓
79	↓
80	↓
81	↓
82	↓
83	↓
84	↓
85	↓
86	↓
87	↓
88	↓
89	↓
90	↓
91	↓
92	↓
93	↓
94	↓
95	AWMID high 7bit
96	AWMID low 7bit
97	AFTMD
98	SPTPNT
99	Refer to 09~12 of Table 1-4. EFLN1EL0
100	EFSDLV0
101	EFSDVL0
102	EFSDSC0
103	WSOURCE1
104	AWMWAV (MSB)
105	AWMWAV (low 7bit)
106	check_sum
107	\$F7

(2) 2AFM & 2AFM & 1AFM_1AWM

	data
0	\$F0
1	\$43
2	\$0N
3	\$7A
4	byte count
5	↓
6	L
7	M
8	-
9	-
10	0
11	0
12	4
13	0
14	V
15	C
16	↓
17	↓
18	↓
19	↓
20	↓
21	↓
22	↓
23	↓
24	↓
25	↓
26	↓
27	↓
28	Memory_type
29	Memory#
30	Memory#
31	Memory#
32	Refer to 00 of Table 1-3. \$01, \$04, \$06, \$08 (ELMODE)
33	Refer to 20~5D of Table 1-10. EFMODE
34	↓
35	↓
36	↓
37	↓
38	↓
39	↓
40	↓
41	↓
42	↓
43	↓
44	↓
45	↓
46	↓
47	↓
48	↓
49	↓
50	↓
51	↓
52	↓
53	↓
54	↓
55	↓
56	↓
57	↓
58	↓
59	↓
60	↓
61	↓
62	↓
63	↓
64	↓
65	↓
66	↓
67	↓
68	↓
69	↓
70	↓
71	↓
72	↓
73	↓
74	↓
75	↓
76	↓
77	↓
78	↓
79	↓
80	↓
81	↓
82	↓
83	↓
84	↓
85	↓
86	↓
87	↓
88	↓
89	↓
90	↓
91	↓
92	↓
93	↓
94	↓
95	AWMID high 7bit
96	AWMID low 7bit
97	AFTMD
98	SPTPNT
99	Refer to 09~12 of Table 1-4. EFLN1EL0
100	EFSDLV0
101	EFSDVL0
102	EFSDSC0
103	Refer to 09~12 of Table 1-4. EFLN1EL1
104	EFSDLV1
105	EFSDVL1
106	EFSDSC1

107	WSOURCE1
108	AWMWAV (MSB)
109	AWMWAV (low 7bit)
110	WSOURCE2
111	AWMWAV (MSB)
112	AWMWAV (low 7bit)
113	check_sum
114	\$F7

(3) 4AFM & 4AWM & 2AFM_2AWM

Note) Memory# ; \$30~\$3F

When receiving, cancel if Memory# is other than the above.

	data
0	\$F0
1	\$43
2	\$0N
3	\$7A
4	byte count
5	↓
6	L
7	M
8	-
9	-
10	0
11	0
12	4
13	0
14	V
15	C
16	↓
17	↓
18	↓
19	↓
20	↓
21	↓
22	↓
23	↓
24	↓
25	↓
26	↓
27	↓
28	Memory_type
29	Memory#
30	Memory#
31	Memory#
32	Refer to 00 of Table 1-3. \$02, \$07, \$09 (ELMODE)
33	Refer to 20~5D of Table 1-10. EFMODE
34	↓
35	↓
36	↓
37	↓
38	↓
39	↓
40	↓
41	↓
42	↓
43	↓
44	↓
45	↓
46	↓
47	↓
48	↓
49	↓
50	↓
51	↓
52	↓
53	↓
54	↓
55	↓
56	↓
57	↓
58	↓
59	↓
60	↓
61	↓
62	↓
63	↓
64	↓
65	↓
66	↓
67	↓
68	↓
69	↓
70	↓
71	↓
72	↓
73	↓
74	↓
75	↓
76	↓
77	↓
78	↓
79	↓
80	↓
81	↓
82	↓
83	↓
84	↓
85	↓
86	↓
87	↓
88	↓
89	↓
90	↓
91	↓
92	↓
93	↓
94	↓
95	AWMID high 7bit
96	AWMID low 7bit
97	AFTMD
98	SPTPNT
99	Refer to 09~12 of Table 1-4. EFLN1EL0
100	EFSDLV0
101	EFSDVL0
102	EFSDSC0
103	Refer to 09~12 of Table 1-4. EFLN1EL1
104	EFSDLV1
105	EFSDVL1
106	EFSDSC1

107	Refer to 09~12 of Table 1-4. EFLN1EL2
108	EFSDLV2
109	EFSDVL2
110	EFSDSC2
111	Refer to 09~12 of Table 1-4. EFLN1EL3
112	EFSDLV3
113	EFSDVL3
114	EFSDSC3
115	WSOURCE1
116	AWMWAV (MSB)
117	AWMWAV (low 7bit)
118	WSOURCE2
119	AWMWAV (MSB)
120	AWMWAV (low 7bit)
121	WSOURCE3
122	AWMWAV (MSB)
123	AWMWAV (low 7bit)
124	WSOURCE4
125	AWMWAV (MSB)
126	AWMWAV (low 7bit)
127	check_sum
128	\$F7

(4) Drum_set

	data
0	\$F0
1	\$43
2	\$0N
3	\$7A
4	byte count
5	L
6	M
7	-
8	-
9	-
10	0
11	0
12	4
13	0
14	V
15	C
16	\$00
29	Memory_type
30	Memory#
31	
32	Refer to 00 of Table 1-3. \$0A (ELMODE)
33	Refer to 20~5D of Table 1-10. EFMODE
95	AWMID high 7bit
96	AWMID low 7bit
97	AFTMD
98	SPTPNT
99	Refer to 07~09 of Table 1-5. WSOURCEC_1
100	AWMWAV (MSB)
101	AWMWAV (low 7bit)
102	EFLN1C_1
103	EFSDVLC_1
104	
459	WSOURCEC_6
460	AWMWAV (MSB)
461	AWMWAV (low 7bit)
462	EFLN1C_6
463	EFSDVLC_6
464	
465	Refer to 07~09 of Table 1-5. ALTE_0
531	ALTB_0
541	EFSDVLB_0
542	ALTC#_6
619	ALTG_6
629	ALTG_6
630	WSOURCEC_0
631	AWMWAV (MSB)
632	AWMWAV (low 7bit)
663	WSOURCEC_6
664	AWMWAV (MSB)
665	AWMWAV (low 7bit)
666	check_sum
667	\$F7

< Table 3-1 >

Multi Bulk Dump

Note) Memory_type internal ; \$00
 preset ; \$02
 Edit Buffer ; \$7F (Used only when transmitting
 from SY99. Memory# is transmitted
 as \$00, ignored when receiving.)

When receiving Bulk dump, if Memory_type is other than \$7F, this is
 processed as Internal.

Note) Memory# \$00~\$0F; INT1~16

Note) When receiving Bulk dump, bits 6~4 of Memory# are ignored.

	data		data		data
0	\$F0	90	Refer to 00~06 of Table 1-2.	0	\$F0
1	\$43	97	OFVCSW_0, OUT*CH_0	1	\$43
2	\$0N	104	OFVCSW_1, OUT*CH_1	2	\$2N
3	\$7A	111	OFVCSW_2, OUT*CH_2	3	\$7A
4	byte count	118	OFVCSW_3, OUT*CH_3	4	L
5	L	125	OFVCSW_4, OUT*CH_4	5	M
6	M	132	OFVCSW_5, OUT*CH_5	6	-
7	-	139	OFVCSW_6, OUT*CH_6	7	-
8	-	146	OFVCSW_7, OUT*CH_7	8	8
9	-	153	OFVCSW_8, OUT*CH_8	9	1
10	8	160	OFVCSW_9, OUT*CH_9	10	0
11	1	167	OFVCSW10, OUT*CH10	11	1
12	0	174	OFVCSW11, OUT*CH11	12	M
13	1	181	OFVCSW12, OUT*CH12	13	U
14	M	188	OFVCSW13, OUT*CH13	14	\$00
15	U	195	OFVCSW14, OUT*CH14	27	Memory type
16	\$00	201	STPAN15	28	Memory#
29	Memory_type	202	check_sum	29	\$F7
30	Memory#	203	\$F7	30	
31					
32	Refer to 00~13 of Table 1-1.				
↓	MNAME0				
51	MNAME19				
52	Refer to Dummy of Table 1-10.				
↓	EFMODE				
80	ST_MIX2				

dump request

	data
0	\$F0
1	\$43
2	\$2N
3	\$7A
4	L
5	M
6	-
7	-
8	8
9	1
10	0
11	1
12	M
13	U
14	\$00
27	Memory type
28	Memory#
29	\$F7
30	

* Memory type = not \$7F
(Edit buffer)

* Memory type =
 not \$7F
 (Edit buffer)

< Table 3-2 >

Additional Multi Bulk Dump

Note) Memory_type internal ; \$00
 preset ; \$02
 Edit Buffer ; \$7F (Used only when transmitting
 from SY99. Memory# is transmitted
 as \$00, ignored when receiving.)

When receiving Bulk dump, if Memory_type is other than \$7F, this is
 processed as Internal.

Note) Memory # \$00~\$0F; INT1~16

Note) When receiving Bulk dump, bits 6~4 of Memory# are ignored.

data	
0	\$F0
1	\$43
2	\$0N
3	\$7A
4	byte count
5	↓
6	L
7	M
8	-
9	-
10	0
11	0
12	4
13	0
14	M
15	U
16	↓
17	\$00
29	Memory_type
31	Memory#
32	↓
94	EFLFOPH

data	
95	Refer to 07~09 of Table 1-2. EFSDMD0
98	EFSDMD1
101	EFSDMD2
104	EFSDMD3
107	EFSDMD4
110	EFSDMD5
113	EFSDMD6
116	EFSDMD7
119	EFSDMD8
122	EFSDMD9
125	EFSDMD10
128	EFSDMD11
131	EFSDMD12
134	EFSDMD13
137	EFSDMD14
140	EFSDMD15
143	check_sum
144	\$F7

data	
0	\$F0
1	\$43
2	\$2N
3	\$7A
4	L
5	M
6	-
7	-
8	0
9	0
10	4
11	0
12	M
13	U
14	↓
15	\$00
27	Memory type
29	Memory#
30	\$F7

* Memory type =
not \$7F
(Edit buffer)

* Memory type =
 not \$7F
 (Edit buffer)

(5) dump request

	data
0	\$F0
1	\$43
2	\$2N
3	\$7A
4	L
5	M
6	-
7	-
8	0
9	0
10	4
11	0
12	V
13	C
14	\$00
27	Memory type
28	Memory#
29	\$F7
30	

* Memory type = not \$7F (edit buffer)

< Table 4 >

Pan Bulk Dump

Note) Memory_type internal ; \$00
 preset ; \$02

When receiving Bulk dump, processed as Internal regardless of the Memory_type.

Note) Memory# \$00~\$1F ; INT1~32

Note) When receiving Bulk dump, bits 6, 5 of Memory# are ignored.

dump request		data	
0	\$F0	0	\$F0
1	\$43	1	\$43
2	\$0N	2	\$2N
3	\$7A	3	\$7A
4	byte count	4	L
5		5	M
6		6	-
7		7	-
8		8	1
9		9	0
10		10	0
11		11	1
12		12	0
13		13	P
14		14	N
15		15	N
16		16	\$00
29		28	Memory type
30	Memory_type	29	Memory#
31	Memory#	30	\$F7
32	Refer to 00~1A of Table 1-12.		
33	PNSCSEL		
58	PNNAM9		
59	check_sum		
60	\$F7		

< Table 6 >

Master Control Bulk Dump

dump request		data	
0	\$F0	58	CONENA1_1
1	\$43	71	MDRNUM1_1
2	\$0N	72	CONENA2_1
3	\$7A	85	MDRNUM2_1
4	byte count	86	CONENA3_1
5		99	MDRNUM3_1
6		100	CONENA4_1
7		113	MDRNUM4_1
8			
9			
10			
11			
12		114	CONENA1_2
13		505	MDRNUM4_8
14			
15		506	CNAM0_1
16		515	CNAM9_1
31	\$00		
32	Refer to Table 1-15.	516	CNAM0_2
33	FILPGMH (MSB)	585	CNAM9_8
34	FILPGMH (low 7 bits)	586	PGM_1
35	FILPGML (MSB)	591	MNUM_1
36	FILPGML (low 7 bits)		
52	FILMVOLH (MSB)	592	PGM_2
53	FILMVOLH (low 7 bits)	599	MNUM_8
54	FILMVOLL (MSB)	633	
55	FILMVOLL (low 7 bits)		
56	reserved	634	check_sum
57	reserved	635	\$F7

< Table 7 >

System Setup Bulk Dump

dump request		data	
0	\$F0	0	\$F0
1	\$43	1	\$43
2	\$0N	2	\$2N
3	\$7A	3	\$7A
4	byte count	4	L
5		5	M
6		6	-
7		7	-
8		8	8
9		9	1
10		10	0
11		11	1
12		12	S
13		13	Y
14		14	\$00
15		29	
16		30	\$F7
31	\$00		
32	Refer to 00~3F of Table 1-16.		
95	reserve		
96	check_sum		
97	\$F7		

< Table 5 >

Micro Tuning Bulk Dump

Note) Memory# \$00~\$01 ; INT1~2

Note) When receiving Bulk dump, bits 6~1 of Memory# are ignored.

dump request		data	
0	\$F0	0	\$F0
1	\$43	1	\$43
2	\$0N	2	\$2N
3	\$7A	3	\$7A
4	byte count	4	L
5		5	M
6		6	-
7		7	-
8		8	1
9		9	0
10		10	0
11		11	1
12		12	0
13		13	M
14		14	T
15		15	T
16		16	\$00
29		27	\$00
30	\$00	28	\$00
31	Memory#	29	Memory#
		30	\$F7
32	Refer to 0000~017E of Table 1-13.		
33	MCTC_2 (high 7bits)		
286	MCTG_8 (high 7bits)		
287	MCTG_8 (low 7bits)		
288	Refer to 0200~0209 of Table 1-13.		
297	MTNAM_1		
298	check_sum		
299	\$F7		

< Table 8-1 >

Squencer Setup Bulk Dump

		dump request	
		data	data
0	\$F0	0	\$F0
1	\$43	1	\$43
2	\$0N	2	\$2N
3	\$7E	3	\$7E
4	byte count	4	L
5		5	M
6	L	6	-
7	M	7	-
8	-	8	8
9	-	9	1
10	8	10	0
11	1	11	1
12	0	12	S
13	1	13	S
14	S	14	\$F7
15	S		
16	QUANTIZE		
17	CLICK SWITCH		
18	CLICK BEAT		
19	reserve		
20	SYNC MODE		
21	REC MIDI CHANNEL		
22	VELOCITY SW.		
23	CONTROL CHANGE SW.		
24	PITCH BEND SW.		
25	PROGRAM CHANGE SW.		
26	AFTER TOUCH SW.		
27	SYSEXCLUSIVE SW.		
28	MIDI CONTROL SW.		
29	EDIT BEAT/CLOCK		
30	ACCENT1		
31	ACCENT2		
32	ACCENT3		
33	ACCENT4		
34	GATE TYPE		
35	check_sum		
36	\$F7		

< Table 9-1 >

Waveform Data Bulk Dump

Note) FROM, TO When Memory_type of \$01 (card), \$02 or \$03 (pre1), or \$04 (pre2) is specified, FROM and TO are both set to a value of \$00.

		data
0	\$F0	
1	\$43	
2	\$0N	
3	\$7A	
4	byte count	
5		
6	L	
7	M	
8	-	
9	-	
10	0	
11	0	
12	4	
13	0	
14	W	
15	V	
16		
↓		\$00
29		
30	Memory_type	
31	Memory#	
32		Refer to 0~9 of Table 1-9.
↓		WVNAME0
39		↓
↓		WVNAME7
40		WVFROM
41		WVTO
42	check_sum	
43	\$F7	

< Table 9-2 >

Sample data Bulk Dump

		data
0	\$F0	
1	\$43	
2	\$0N	
3	\$7A	
4	byte count	
5		
6	L	
7	M	
8	-	
9	-	
10	0	
11	0	
12	4	
13	0	
14	S	
15	A	
16		
↓		\$00
30		
31	Memory#	
32		Refer to 6~13 of Table 1-17.
↓		SANAM0
39		↓
↓		SANAM7
40	HIKEYCODEH (MSB)	
41	HIKEYCODEH (low 7bit)	
42	HIKEYCODEL (MSB)	
43	HIKEYCODEL (low 7bit)	
44	ORIKEY	
45	PITCHCODEH (MSB)	
46	PITCHCODEH (low 7bit)	
47	PITCHCODEL (MSB)	
48	PITCHCODEL (low 7bit)	
49	LOOPMODE	
50	VOLCODE (MSB)	
51	VOLCODE (low 7bit)	
52	LOWKEYCODEH (MSB)	
53	LOWKEYCODEH (low 7bit)	
54	LOWKEYCODEL (MSB)	
55	LOWKEYCODEL (low 7bit)	
56	check_sum	
57	\$F7	

< Table 8-2 >

Squencer Song & Pattern (KSEQ, NSEQ) Bulk Dump

KSEQ and NSEQ data is converted from 1 byte to 2 byte ascii data and transmitted. The data for one song consists of more than one track of data, where each track begins with F0 0n, (n=track number) and ends with F2. Empty tracks will not be included.

hex	description
F0 00	top of record track #1
---	time/event/control data
F2	end of record track #1
---	track #2~#15 data
F0 0F	top of record track #16
---	time/event/control data
F2	end of record track #16

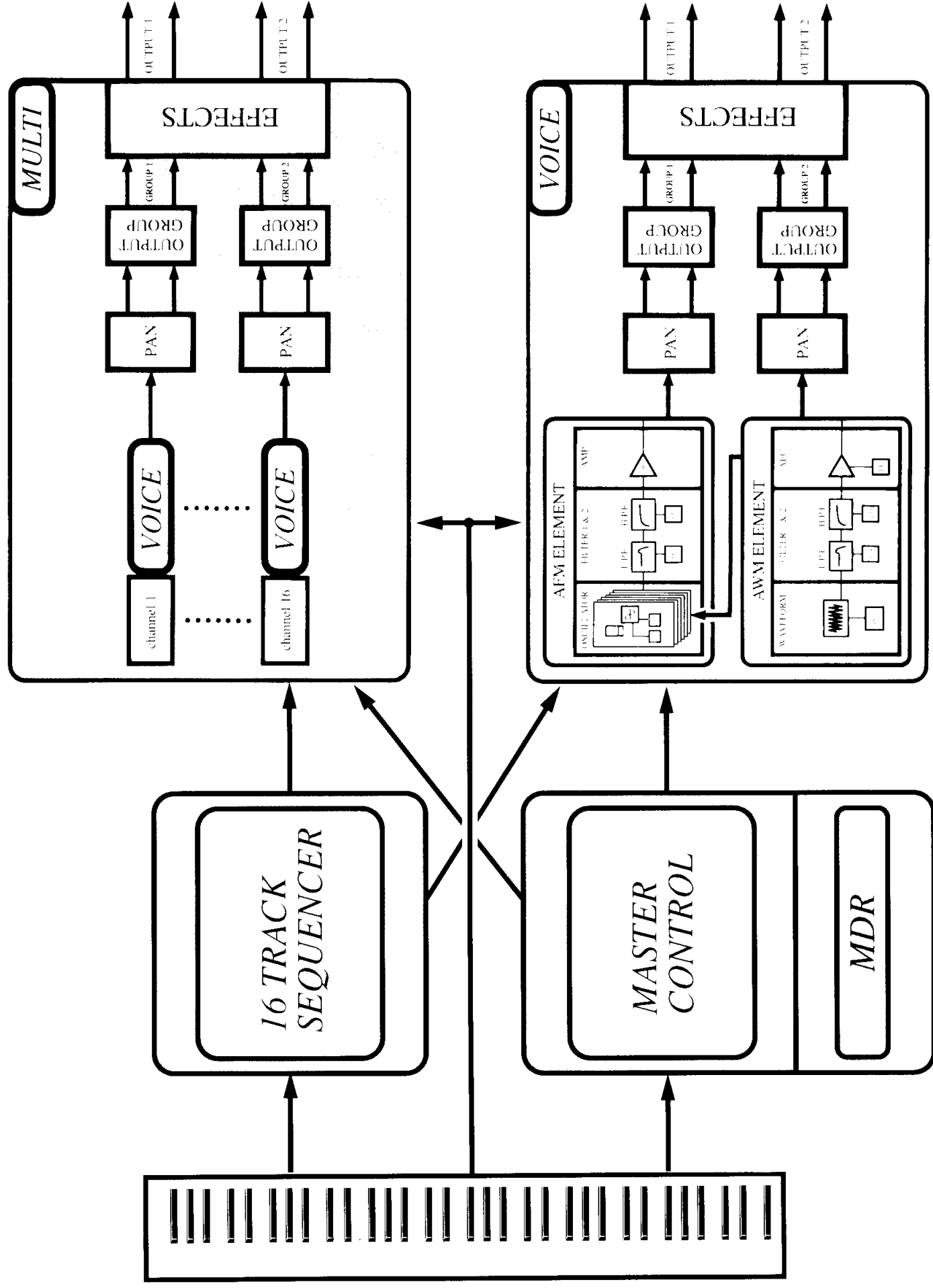
Function ...		Transmitted	Recognized	Remarks
Basic Channel	Default	1 - 16	1 - 16	memorized
	Changed	1 - 16	1 - 16	
Mode	Default	3	1,2,3,4	memorized
	Messages	x	x	
	Altered	*****	x	
Note Number : True voice		28 - 103 *****	0 - 127 1 - 127	
Velocity	Note ON	o 9nH,v=1-127	o v=1-127	
	Note OFF	x 9nH,v=0	x	
After Touch	Key's	x	x	
	Ch's	o	o	
Pitch Bender		o	o 0-12 semi	7 bit resolution
Control Change	0	o	o	Bank select
	1	o M.Wheel	o	
	2	o Breath cont.	o	
	4	o Foot cont.	o	
	6	o Data entry	o	
	7	o Foot volume	o	Volume
	32	o	o	Bank select
	64	o Sustain sw.	o	Sustain
	65	o Portamento sw	o	Portamento
	96	o inc.	o	
	97	o dec.	o	
	1 - 6	o Assignable	o	
	8 - 120	o Assignable	o	
Prog Change : True #		o 0-79 *1 *****	o 0-79 *2	
System Exclusive		o	o	voice etc.
Common : Song Pos : Song Sel : Tune		See the sequencer part.		
System :Clock Real Time :Commands				
Aux	:Local ON/OFF	x	x	
	:All Notes OFF	x	x	
Mes-	:Active Sense	o	o	
	sages:Reset	x	x	
Note *1 ; Send PC,Master control: 0 - 127 *2 ; voice : 0 - 63 , multi : 64 - 79				
Mode 1 : OMNI ON, POLY Mode 2 : OMNI ON, MONO o : Yes Mode 3 : OMNI OFF, POLY Mode 4 : OMNI OFF, MONO x : No				

Function ...		Transmitted	Recognized	Remarks
Basic	Default	1 - 16	1 - 16	memorized
Channel	Changed	1 - 16	1 - 16	
Mode	Default	x	x	
	Messages	x	x	
	Altered	*****	x	
Note		0 - 127	0 - 127	
Number : True voice		*****		
Velocity	Note ON	o 9nH,v=1-127	o v=1-127	
	Note OFF	x 9nH,v=0	x	
After	Key's	x	x	
Touch	Ch's	o	o	
Pitch Bender		o	o	
0 - 120		o	o	
Control Change				
Prog		o 0 - 127	o 0 - 127	
Change : True #		*****		
System Exclusive		o	o	Song data etc.
	: Song Pos	o	o	
Common	: Song Sel	o	o	
	: Tune	x	x	
System	:Clock	o	o	
Real Time	:Commands	o	o	
Aux	:Local ON/OFF	x	x	
	:All Notes OFF	x	x	
Mes-	:Active Sense	x	x	
sages:	Reset	x	x	
Notes				

Mode 1 : OMNI ON, POLY
 Mode 3 : OMNI OFF, POLY

Mode 2 : OMNI ON, MONO
 Mode 4 : OMNI OFF, MONO

o : Yes
 x : No



PRESET/INTERNAL VOICE

PRESET 1 (64 voices)

#	Bank A	Bank B	Bank C	Bank D
1	AP Rocks	SP Alaska	BR TmoSec	ST Concert
2	APICsRock	SP SawPad	BR SigBano	ST Chestra
3	AP Concert	SP Square	BR JazzT+P	BR Soltz
4	AP Sig-ay	SP ELEGANT	BR MuteT+P	ME BgNeck
5	EP T6Stage	SP Dig Pad	BR F+Horns	PC Snapper
6	EP Classic	SP LasRed	BR DrkHorn	PC Marimba
7	EP NiteHw	SP Sweeper	BR Azen '6	PC Vibes
8	EP Berose	SP Flash	BR DaBulps	PC MusCBx
9	EP BellRing	SP HrosiPd	AR Spaz	PC T+ant
10	EP Dixism	SP Skware	BR P+moed	PC G ocht
11	EP GrnQual	SPA peggi	BR Sigi-ay	PC Bala+
12	EP VoxLayr	SP Veckar	ST Octaves	PC Berim
13	KY Smokey	SP Crystal	ST ChoAra	SE Siltner
14	KY CscClav	SP T+works	ST Rosin	DR Aids
15	KV Cavit	SP Polyodr	ST Quartz	DR Perc
16	KV ResoCliv	SP WarmJet	ST P+zza	DR Mixcd

PRESET 2 (64 voices)

#	Bank A	Bank B	Bank C	Bank D
1	SC Heretic	PL Steel6	BA Pickad	WN Tenny
2	SC AeroPro	PL JazzGr	BA Sapped	WN SaxSeul
3	SC Jupite	PL Nylon6	BA Frigs	WN Alto
4	SC RezWhab	PL 12String	BA Frutes	WN Soprano
5	SC Plectra	PL Eko 12S	BA Classic	WN Carve
6	SC Quata	PL Echoes6	BA Uprght	WN PanPipe
7	SC Plastic	PL Caster	BA DXS ap	ME Phantom
8	SC Tanager	PL SoloLead	BA Arabass	ME 5HsMan
9	SC Gizmo	PL RockAI	BA ResoSyr	ME 12mpor
10	SC Healing	SL SawLead	BA FalSyr	MES Glooob
11	SC Angelic	SL EchoSaw	BA Mogue	MF Asa a
12	CH Gasine	SL DLux	OR RJazzy	ME Dreams
13	CH Italian	SL Sync	OR BoozAB	ME Galaxy
14	CH Vespers	SL Square	OR Droop	ME ISS
15	CH Nebula	SL PulseWM	OR Polio	MF 7sod
16	CH Witches	SL Lye	OR Silica	ME TricSlar

INTERNAL (64 voices)

#	Bank A	Bank B	Bank C	Bank D
1	SP Elena	AP Bright	BA Extras	WN Hero A's
2	SP Dreamod	EP Be EP	BA Proxy	WN Hrd Tenr
3	SP Freeze	EP Propion	BA Rode	WN Bar-Sax
4	SP Polygar	EP DualDA	BA Ve Slap	WN AmbHarp
5	SP DarkPac	OR Ghosty	BA S're	SP MoonPac
6	SP Dig82	KV S'reeze	BA Ubrigt	ME Cosmos
7	SP Digma	SL P'taSaw	BA Sarcous	ME A'rola
8	SP SynStr	SL OcSoc	BA DigW c	ME Ga xy
9	SC Magic	ST S'rgPac	PL E'k'k	ME Caris's
10	SC DrzSib	ST ClasStr	PL M'at'ed	SE Astral
11	SC SapCv	ST Tremolo	PL O'Dr've	KV Harps:
12	SC Anagry	ST Ok Syn's	PL Stratus	BR Fa
13	SC Steps	ST Viol'n	PL E'Mute	PL Ve Gr
14	SC DigSib	ST Cel'o	PL Ve M're	KS Ang -2
15	Ch'Cr'orWn	BR HouseAT	PL Harp	KS Pac Sax
16	Ch'OccAr	BR S'zSwe	PL LAPzz	KS LapCor'o

The original internal voices can be recoded by loading the file named "N-VOICE" from the "Chick Corea demo disk."

SP	WN	PL	SE	Sound Effects
Syn Pad	Winds	P	DR	Drum/Vocals
Acoustic Piano	ST	KY		
Bass	BA	OR		
Electric Piano	SC	PC		
Musical Effect	SL	CH		

the two characters of pres. voice names indicate the type of voice

Element	AWM $\times 1$ or AFW $\times 1$
2 elements	AWM $\times 2$ or AFW $\times 2$
3 elements	AWM $\times 3$ or AFW $\times 3$
4 elements	AWM $\times 4$ or AFW $\times 4$ or AWM $\times 2$ or AFW $\times 2$

The third character of these voice names indicates the number of glimmers used by the voice.

PRESET MULTI

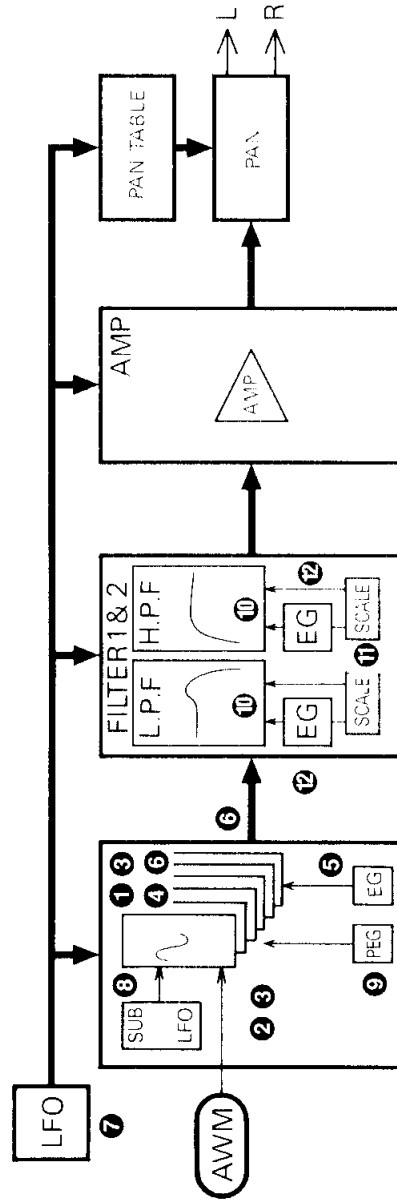
No.	Multi Name	Voice Number									
		01	02	03	04	05	06	07	08	09	10
01	Popular Tune	EP Gtr Duo BR Big Band KY Ctr Chv EP Ctr Chv	BA Picked WN Tenor BA Wn Tenor off	AP Siglav Off toon AP Rocks off	EP Siglav Off toon S. Snc Sam off	PL Ctr Bass off	PLI 2 Sing off	SP Legat off	S. Octaves DR K's S. Snc DR K's		
02	Funky Tune	EP Ctr Chv off	BA Wn Tenor off	AP Rocks off	EP Big Brn off	PL Ctr Bass off	BR Big Band off	BR Big Band off	S. Snc DR K's		
03	Ballade	EP Ctr Chv WN Soprano	BA Ctr Chv WN Pano	AP Siglav S. Snc Sam	SC Hea CH Gas re	PL 2 Sing off	SP Wn Tenor off	SP Square DR Perc	S. Snc DR K's		
04	House	SC Plastik off	BA Moque off	AP Ctr Rock off	SC Hea CH Gas re	PL 2 Sing off	SP Wn Tenor off	BR Big Band off	S. Snc DR K's		
05	Standard Rock	EP Ctr Chv off	BA Picked off	AP Rocks off	PL Ctr Bass off	PL Ctr Bass off	SP Wn Tenor off	BR Big Band off	S. Snc DR K's		
06	American Rock	OR Deep WN Tenor	BA Samped off	AP Rocks off	EP Dtr off	KY Ctr Bass off	SP Wn Tenor off	BR Big Band off	S. Snc DR K's		
07	Combo Jazz	AP Concert off	BA Jorg off	PL Jazz Gtr off	WN Alto off	off	off	off	off		
08	2 Horn Jazz Quintet	PL Jazz Gtr off	BA Jorg off	AP Rocks off	WN Alto off	WN Soprano off	WN Soprano off	BR Big Band off	S. Snc DR K's		

[illegible]

AFM Element Edit Reference

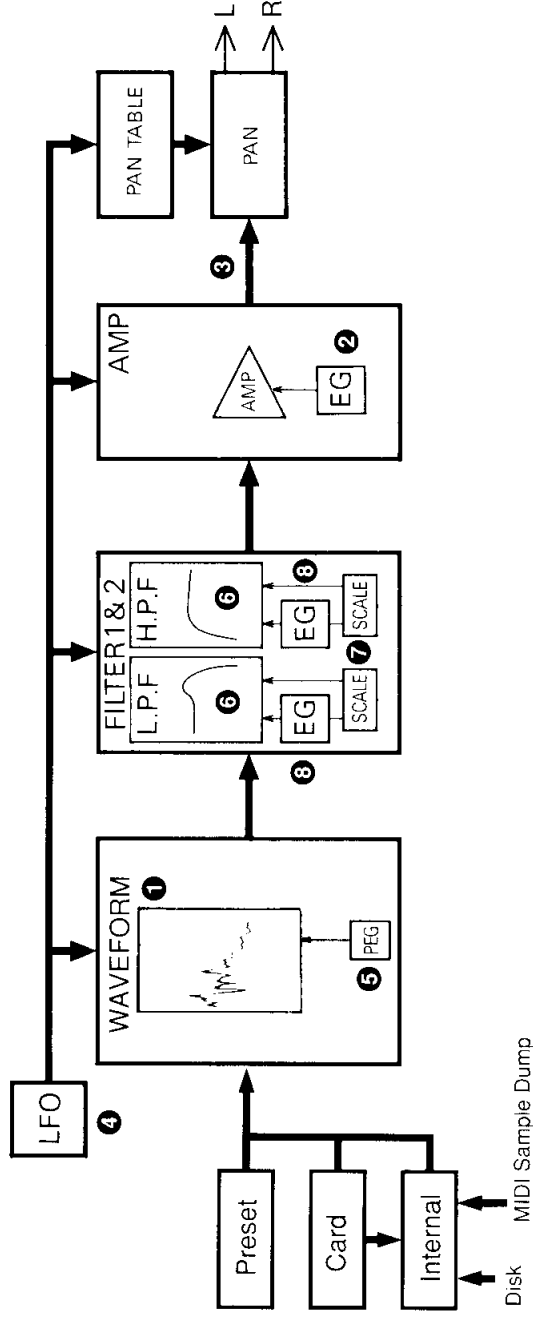
AFM element edit jobs

1. Algorithm
- ① F1 (Form) #234
 ② F2 (Exir) #231
 ③ F3 (Inpt) #232
 ④ Oscillator #233
 ⑤ AFM EG #235
 ⑥ AFM operator output #236...240
 ⑦ AFM sensitivity #241...242
 ⑧ AFM LFO #243
 ⑨ F1 (Main) #244
 ⑩ F2 (Sub) #245
 ⑪ AFM pitch EG #246...247
 ⑫ AFM filter #248
 ⑬ F1 Cutoff frequency #249
 ⑭ F2 Cutoff scaling #250...251
 ⑮ F3 Cutoff EG #253...255



One carrier					Two carriers					Three carriers				
⑦	⑧	⑨	⑩	⑪	⑫	⑬	⑭	⑮	⑯	⑰	⑱	⑲	⑳	㉑
01	02	03	04	05	06	07	08	09	10	11	12	13	14	15
16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
Four carriers					Four carriers					Four carriers				
46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
61	62	63	64	65	66	67	68	69	70	71	72	73	74	75
76	77	78	79	80	81	82	83	84	85	86	87	88	89	90
91	92	93	94	95	96	97	98	99	100	101	102	103	104	105

AWM element edit jobs



- 1** AWM Waveform Set #257
- 2** AWM Amplitude EG #258
- 3** AWM Output #259
- 4** AWM Sensitivity #260
- 5** AWM LFO #261
- 6** AWM Pitch EG #262...263
- 7** AWM Filter Set #264
- 8** F1 Cutoff frequency #265
- 9** F2 Cutoff scaling #266...267
- 10** F3 Cutoff EG #268...271

AWM2 Waveform list

Preset 1

1	Piano	40	Celestia	79	EG Harm1	118	Ride	1	Piano Nb	29	Typist	57	DigiVox2	85	Snuff 20
2	Trumpet	41	Harp	80	EG Harm2	119	Croak	2	E.P. Np	30	Belting	58	DigiVox3	86	Snuff 21
3	MuteTp 1	42	Pipe Wv	81	EG Mute	120	Claps 1	3	Vibe Np	31	SeqLatin	59	DigiVox4	87	Snuff 22
4	MuteTp 2	43	AngBrS1	82	EG Comp	121	Claps 2	4	DmpPiano	32	Elemagic	60	DigiVox5	88	Snuff 23
5	Horn	44	AngBrS2	83	EG Dist	122	Cowbell1	5	Bottle 1	33	Vox Bell	61	Pulse 10	89	Snuff 24
6	Flugel	45	AngBrS3	84	EG Pluk1	123	Cowbell2	6	Bottle 2	34	Melow	62	Pulse 25	90	Snuff 25
7	Trombone	46	Pad 1	85	EG Pluk2	124	Tambor	7	Bottle 3	35	BigSyn L	63	Pulse 50	91	Snuff 26
8	Tuba	47	Pad 2	86	BD 1	125	Shaker	8	Tube	36	BigSyn R	64	Tri	92	Snuff 27
9	Brass	48	AngBrS4	87	BD 2	126	FngSnao	9	Vocal Ga	37	VoxGrace	65	DigiWild	93	Snuff 28
10	BisFall	49	FrisSyn	88	BD 3	127	AngPerc	10	Vocal Ba	38	Cry Bell	66	Snuff 1	94	Snuff 29
11	Tenor1	50	Chorus	89	BD 4	128	NoisePrc	11	Sax Tran	39	Voices	67	Snuff 2	95	Snuff 30
12	Tenor2	51	Chorus L	90	BD 5	129	Scratch	12	Bow Tran	40	AngSaw1	68	Snuff 3	96	Snuff 31
13	Alto Sax	52	Chorus R	91	BD 6	130	Agogo	13	Blub	41	AngSaw2	69	Snuff 4	97	Snuff 32
14	Bartone	53	Itopia	92	BD 7	131	Berimbau	14	Tear	42	CS Saw	70	Snuff 5	98	Snuff 33
15	Soprano	54	Choir	93	BD 8	132	Bongo	15	Bamboo	43	CS Sqr	71	Snuff 6	99	Snuff 34
16	Tenors	55	ConChor	94	SD 1	133	Cabasa	16	Cup Echo	44	Digital1	72	Snuff 7	100	Snuff 35
17	Flute	56	Vibe	95	SD 2	134	Cga Ht	17	DigiAlk	45	Digital2	73	Snuff 8	101	Snuff 36
18	Clarinet	57	Marimba	96	SD 3	135	CgaHSlp	18	Temp Ra	46	Digital3	74	Snuff 9	102	Snuff 37
19	Piccolo	58	Tubular	97	SD 4	136	Cga Lo	19	Giri	47	Digital4	75	Snuff 10	103	Snuff 38
20	Reed Wv	59	Xylophon	98	SD 5	137	CgaLoSlp	20	Water	48	Digital5	76	Snuff 11	104	Snuff 39
21	Bassoon	60	Glocken	99	SD 6	138	Clave	21	Steam	49	Digital6	77	Snuff 12	105	Snuff 40
22	Recorder	61	SteelDrum	100	SD 7	139	Guro 1	22	Narrow	50	Digital7	78	Snuff 13	106	Snuff 41
23	MtReedWv	62	HandBell	101	SD 8	140	Guro 2	23	Airy	51	Digital8	79	Snuff 14	107	Snuff 42
24	PanFlute	63	Shamisen	102	SD 9	141	Maracas	24	Styoli	52	Digital9	80	Snuff 15	108	Snuff 43
25	Violin	64	Koto	103	SD Side	142	SD roll	25	Noise	53	Digital10	81	Snuff 16	109	Snuff 44
26	Cello	65	Harp	104	Tom 1	143	Tabla Hi	26	Bell Mix	54	Digital11	82	Snuff 17	110	Snuff 45
27	ContraBs	66	Sitar	105	Tom 2	144	Tabla Lo	27	Haaa	55	Digital12	83	Snuff 18	111	Snuff 46
28	Pizz	67	E Bass 1	106	Tom 3	145	Temple	28	OnAttack	56	DigiVox1	84	Snuff 19	112	Snuff 47
29	SectPizz	68	E Bass 2	107	Tom 4	146	Timbale1								
30	Strings1	69	E Bass 3	108	Tom 5	147	Timbale2								
31	Strings2	70	TrmbBass	109	Tom 6	148	Timbani								
32	Strings3	71	SlapBass	110	HH foot	149	Whistle								
33	StringsR	72	Fretless	111	HH light	150	Belltree								
34	Organ 1	73	WoodBass	112	HH mid	151	BDs Wv								
35	Organ 2	74	GrSteel	113	HH heavy	152	SDs Wv								
36	E.P.Wv1	75	GrNylon	114	HH open	153	Toms Wv								
37	E.P.Wv2	76	12String	115	HHAnlg	154	CymbalWv								
38	Clavi 1	77	EG Sng1	116	HHopAnrg	155	Drums Wv								
39	Clavi 2	78	EG Humbk	117	Crash										

Preset 2

1	Piano Nb	29	Typist	57	DigiVox2	85	Snuff 20
2	E.P. Np	30	Belting	58	DigiVox3	86	Snuff 21
3	Vibe Np	31	SeqLatin	59	DigiVox4	87	Snuff 22
4	DmpPiano	32	Elemagic	60	DigiVox5	88	Snuff 23
5	Bottle 1	33	Vox Bell	61	Pulse 10	89	Snuff 24
6	Bottle 2	34	Melow	62	Pulse 25	90	Snuff 25
7	Bottle 3	35	BigSyn L	63	Pulse 50	91	Snuff 26
8	Tube	36	BigSyn R	64	Tri	92	Snuff 27
9	Vocal Ga	37	VoxGrace	65	DigiWild	93	Snuff 28
10	Vocal Ba	38	Cry Bell	66	Snuff 1	94	Snuff 29
11	Sax Tran	39	Voices	67	Snuff 2	95	Snuff 30
12	Bow Tran	40	AngSaw1	68	Snuff 3	96	Snuff 31
13	Blub	41	AngSaw2	69	Snuff 4	97	Snuff 32
14	Tear	42	CS Saw	70	Snuff 5	98	Snuff 33
15	Bamboo	43	CS Sqr	71	Snuff 6	99	Snuff 34
16	Cup Echo	44	Digital1	72	Snuff 7	100	Snuff 35
17	DigiAlk	45	Digital2	73	Snuff 8	101	Snuff 36
18	Temp Ra	46	Digital3	74	Snuff 9	102	Snuff 37
19	Giri	47	Digital4	75	Snuff 10	103	Snuff 38
20	Water	48	Digital5	76	Snuff 11	104	Snuff 39
21	Steam	49	Digital6	77	Snuff 12	105	Snuff 40
22	Narrow	50	Digital7	78	Snuff 13	106	Snuff 41
23	Airy	51	Digital8	79	Snuff 14	107	Snuff 42
24	Styoli	52	Digital9	80	Snuff 15	108	Snuff 43
25	Noise	53	Digital10	81	Snuff 16	109	Snuff 44
26	Bell Mix	54	Digital11	82	Snuff 17	110	Snuff 45
27	Haaa	55	Digital12	83	Snuff 18	111	Snuff 46
28	OnAttack	56	DigiVox1	84	Snuff 19	112	Snuff 47

VOICE PLAY MODE		page #100
Voice Directory	Controller View	#102
VOICE EDIT MODE		
Voice Mode	#200	#230
Voice Common Data		
01. Element level	#201	#231
02. Element detune	#202	#232
03. Element note shift	#203	#233
04. Element note limit	#204	#234
05. Element velocity limit	#205	#235
06. Element dynamic pan	#206	#236
07. Output group select	#207	#237
08. Random pitch	#208	#238
09. Portamento	#209	#239
10. Effect set	#210	#240
01. Effect mode select	#211	#241
02. Effect send	#212	#242
03. Effect 1 parameter	#213	#243
04. Effect 2 parameter	#214	#244
05. Effect control	#215	#245
06. Effect 1 parameter	#216	#246
07. Effect 2 parameter	#217	#247
08. Effect control	#218	#248
11. Micro tuning set	#219	#249
01. Micro tuning edit	#220	#250
02. Micro tuning data	#221	#251
03. Micro tuning name	#222	#252
12. Controller set	#223	#253
01. Controller set	#224	#254
02. Controller set	#225	#255
03. Controller set	#226	#256
04. Controller set	#227	#257
05. Controller set	#228	#258
13. Voice name	#229	#259
14. Initialize voice	#230	#260
15. Recall voice	#231	#261
Drum Set Data		#262
01. Voice volume	#263	#264
02. Wave data set	#264	#265
03. Effect set	#265	#266
04. Controller set	#266	#267
05. Voice name	#267	#268
07. Initialize voice	#268	#269
08. Recall voice	#269	#270
	#270	#271
	#271	#272
	#272	#273
	#273	#274
	#274	#275
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	#366	#367
	#367	#368
	#368	#369
	#369	#370
	#370	#3

SONG MODE

Song Setup	page #622	Song Edit	page #606
01. Receive event	#623	Graphic	—
02. Midi control	#624	Data	—
03. Accent level	#625	Insert	—
04. Clock/Beat	#626	Change	—
Transmit Channel	#627	Delete	—
Song Name	#628	Song Edit Job 1	page #600
Song Directory	#629	01. Append song	#601
		02. Cut song	#602
		03. Copy song	#603
		04. Copy track	#604
		05. Clear song	#605

Song Edit Job 2	page #607
01. Quantize	#607
02. Modify gate time	#608
03. Modify velocity	#609
04. Crescendo	#610
05. Transpose	#611
06. Thin out	#612
07. Erase event	#613
08. Note shifts	#614
09. Move clock	#615
10. Copy measure	#616
11. Erase measure	#617
12. Delete measure	#618
13. Create measure	#619
14. Mix track	#620
15. Erase track	#621

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